

REVIEW

Open Access



A systematic literature review of machine learning and deep learning for urban heat island modelling

Nida Samad^{1*}, M. Farnaghi^{1†}, F. O. Ostermann^{1†} and R. Zurita-Milla^{1†}

[†]M. Farnaghi, F. O. Ostermann and R. Zurita-Milla contributed equally to this work.

*Correspondence:

Nida Samad

n.samad@utwente.nl

[†]Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede, the Netherlands

Abstract

The urban heat island (UHI) effect is a growing environmental and public-health challenge intensified by climate change and rapid urbanisation. While physical models have traditionally been used to study UHI processes, they often struggle to represent the complex spatial–temporal structure of urban environments. Recent advances in machine learning (ML) and deep learning (DL) offer new opportunities to address these limitations; however, their application in UHI research remains limited. This systematic literature review analyses 85 peer-reviewed studies published between June 2019 and December 2024, structured around four research questions addressing model typology, spatial–temporal scale, evaluation and explainability practices, and remaining research gaps. Adopting a model-centric perspective, this review analyses how different ML and DL model classes are designed, evaluated, and applied to represent multi-scale UHI processes, rather than treating UHI modelling as a purely descriptive comparison of algorithms. The results show that static spatial snapshot models remain dominant, particularly for surface-level classification and mapping tasks, while spatio-temporal and hybrid DL architectures are increasingly applied for short-term forecasting and dynamic UHI analysis. Most studies focus on single-city case studies, with limited evidence of cross-city or cross-climate generalisation. Regarding evaluation practices, only nine of the 85 studies incorporated physics-informed or hybrid approaches, and the adoption of XAI techniques remained limited. This review identifies critical methodological gaps and highlights the need for transferable and generalisable models, robust spatio-temporal evaluation strategies, and integrated explainability frameworks to move ML- and DL-based UHI modelling beyond experimental case studies toward reliable, interpretable, and decision-ready tools for urban planning and heat mitigation.

Keywords Urban heat island, Machine learning, Deep learning, Explainable artificial intelligence, Systematic literature review

1 Introduction

Cities worldwide are facing growing challenges due to climate change, and the urban heat island (UHI) effect has emerged as a critical issue [1]. UHIs are areas that experience higher temperatures than their surrounding areas owing to factors such as urban



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

morphology, less vegetation, and anthropogenic heat [2]. UHIs lead to higher public health risks [3], a significant increase in the energy demand for cooling, and other direct and indirect effects on the environment. Such persistently high temperatures threaten the sustainability of life in cities [4]. Thus, UHI modelling is needed to assist in mitigating the UHI effects and support climate-resilient urban planning.

Historically, UHI research has relied on physical modelling to simulate heat fluxes. Although robust, physical modelling often requires substantial computational resources and fails to adapt to rapidly changing urban morphology and scales [5]. Accurate simulations of heat distribution in urban areas require careful consideration of the horizontal and vertical scales. Horizontal scales represent the spatial extent of the study area, typically at three levels: microscale (less than 100 m), local (100m to 1–2 km), and meso (greater than 2 km) [6]. Vertically, heat is measured in different layers: the surface layer (ground level), canopy layer (from the ground to building rooftops), and boundary layer (higher up in the atmosphere) [7, 8]. Thus, surface urban heat islands (SUHI) measure heat only at the ground level, while canopy layer urban heat islands (CLUHI) measure heat from the ground up to building height [9]. The complexity of UHI modelling is increased by temporal changes and variations across scales, affecting the accuracy of heat distribution simulations. Traditional physical models do not effectively address these complexities, making them insufficient for accurately capturing the dynamic nature of urban heat [5, 10]. Moreover, many existing frameworks still lack systematic validation and reproducibility protocols, limiting their transparency and hindering meaningful cross-city comparisons.

Given the limitations of traditional models in addressing the multidimensional complexities of the UHI effect, the growing threats posed by these effects necessitate the development of more sophisticated modelling approaches. Thus, the need for more flexible and accurate models to identify UHI effects has driven a recent shift in UHI modelling from physical to Machine Learning (ML) models. Over time, ML models have become increasingly popular because of their accuracy and capacity to identify hidden patterns from vast and diverse datasets [11, 12]. Being versatile, ML models can handle data, including numerical and categorical data, satellite imagery, locally sensed environmental variables, and complex formats and modalities [13, 14]. This accuracy and versatility aid in the development of effective mitigation strategies [15, 16]. However, as UHI dynamics become increasingly complex and datasets grow larger, there is an increasing demand for more accurate modelling [17]. Deep learning (DL) models, as a subset of ML techniques, are capable of handling even more complex and larger amounts of data and, therefore, have been increasingly used in many applications recently [18]. These improvements have facilitated the development of more complex models that can not only process extensive datasets and uncover intricate patterns but also handle hidden relationships of scales [19]. However, these advanced models also introduce new challenges, including high computational costs, limited interpretability, and inconsistent reporting of model performance and uncertainty. Addressing these aspects is crucial to ensure that DL models remain reliable and practically applicable in urban climate analyses. In this review, “ML” refers to architectures such as Random Forest (RF), Support Vector Machine (SVM), Gradient Boosting, Multiple Linear Regression (MLR), and Support Vector Regression (SVR), whereas “DL” refers to multi-layer neural architectures such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks

(LSTMs), Gated Recurrent Units (GRUs), Transformers, and Generative Adversarial Networks (GANs).

In recent UHI studies, ML and DL have been applied to address two fundamental but distinct modelling tasks: the classification of high-risk areas based on heat intensity [20] and the forecasting of future UHI patterns [19, 21]. These tasks serve distinct and at the same time complementary purposes. Classification helps planners and decision-makers understand which parts of a city are most vulnerable and where structures, such as greening or shading, are urgently needed [20, 22]. It provides a view based on existing spatial characteristics and offers a foundation for risk assessment and targeted planning. In contrast, prediction focuses on the future. It involves estimating the temperature or heat intensity over time, which is critical for early warning systems, heatwave preparedness, and long-term urban climate planning [23]. Predictive approaches use historical and environmental data to anticipate how heat will evolve in specific locations, allowing cities to take preventive action before extreme events occur. These two tasks serve complementary purposes and require different model architectures and data inputs. As the use of ML and DL in UHI modelling continues to expand across these distinct tasks, it is becoming increasingly important to assess how effectively these models address the unique challenges of spatial and temporal UHI analyses.

Considering the rapid developments in the ML and DL domains, there is a constant need to systematically evaluate how recent ML and DL models reshape UHI modelling practices and outcomes. A systematic literature review (SLR) is a suitable approach to keep up with the growing use of ML and DL in UHI research. Previously, Deilami et al. [24] reviewed research from 1965 to 2017, covering spatio-temporal factors that influence UHI, and how satellite data and state-of-the-art ML and DL methods were taking the lead in UHI studies. Later, a review by Kim and Brown [25] provided a detailed examination of UHI intensity and magnitude estimation from studies conducted from 1972 to 2019. This study highlighted critical gaps in the methodologies for UHI modelling, especially the need for models that can address the urban multi-scale complexities. Their work facilitated an understanding of the challenges in estimating the UHI but focused mainly on traditional methods and did not fully explore the possibilities offered by new computational approaches, such as ML and DL approaches. Our review builds directly on this body of work by focusing on peer-reviewed studies published from June 2019 onward, thereby capturing recent advances in ML and DL model architectures, data scalability, and spatio-temporal representation that were not covered in earlier reviews. More importantly, unlike existing reviews that primarily summarise UHI drivers, estimation methods, or report ML and DL applications descriptively, this review adopts a model-centric perspective. It systematically examines how different ML and DL model classes are designed, evaluated, and applied to represent multi-scale UHI dynamics across spatial and temporal dimensions. By analysing model typology, scale representation, evaluation practices, and the integration of explainability and hybrid approaches, this review goes beyond a descriptive inventory of algorithms and provides a structured assessment of the capabilities and limitations of contemporary ML and DL models for robust and transferable UHI modelling. Thus, it offers a distinct and updated contribution to the literature, supporting both methodological advancement and decision-ready urban heat mitigation strategies.

This review makes three major contributions. First, we present a model-centric synthesis that connects model architecture, spatial-temporal scales, and application types in UHI research. Second, we integrate methodological perspectives on performance evaluation and explainability to identify where existing ML and DL frameworks succeed or fall short. Third, it provides a forward-looking discussion of research gaps and emerging directions for creating interpretable, scalable, and reproducible ML and DL-based UHI models. Together, these contributions offer an updated understanding of how data-driven approaches are transforming UHI modelling and how future research can enhance their scientific and practical value.

This perspective emphasises three core aspects that are necessary for the effective use of ML and DL in UHI modelling.

1. **Model typology** – the classification of modelling approaches according to their architectural and methodological characteristics [26, 27]. A clear typology enables researchers to assess whether specific models are suitable for capturing the complex spatial, temporal scales dynamics of UHI [23, 28] and then select appropriate models [29].
2. **Performance evaluation** – a systematic evaluation of how ML and DL based UHI models perform and generalise [26, 30].
3. **Explainability and interpretability** – essential for decision support in urban planning [31]. Understanding model reasoning enhances trust and applicability [32].

Therefore, the objective of this review is to provide a model-centric review of ML and DL in the context of the UHI to identify research gaps and offer guidance for future studies. The four main research questions for this SLR are as follows:

- **RQ1:** What ML and DL model typologies are used for UHI classification and prediction at different horizontal and vertical scales?
- **RQ2:** Which computational environments, programming languages, libraries, or software tools were used to implement ML and DL models for UHI classification and prediction?
- **RQ3:** What was the performance of ML and DL models used in UHI studies in terms of accuracy, generalisability, and computational efficiency?
- **RQ4:** What explainability techniques have been used in ML and DL-based UHI modelling?

To answer these questions, we conducted an SLR of studies published between 1 June 2019 and 31 December 2024. The next section describes the SLR design details. Section 3 presents the results, followed by a discussion in Sect. 4 that relates the findings to the research questions of this study. Section 5 identifies research gaps for future work to address, and Sect. 6 concludes the article by synthesising key insights and outlining the implications for future research and practice.

2 Methodology

2.1 Review framework

This SLR followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines to ensure methodological transparency and reproducibility [33].

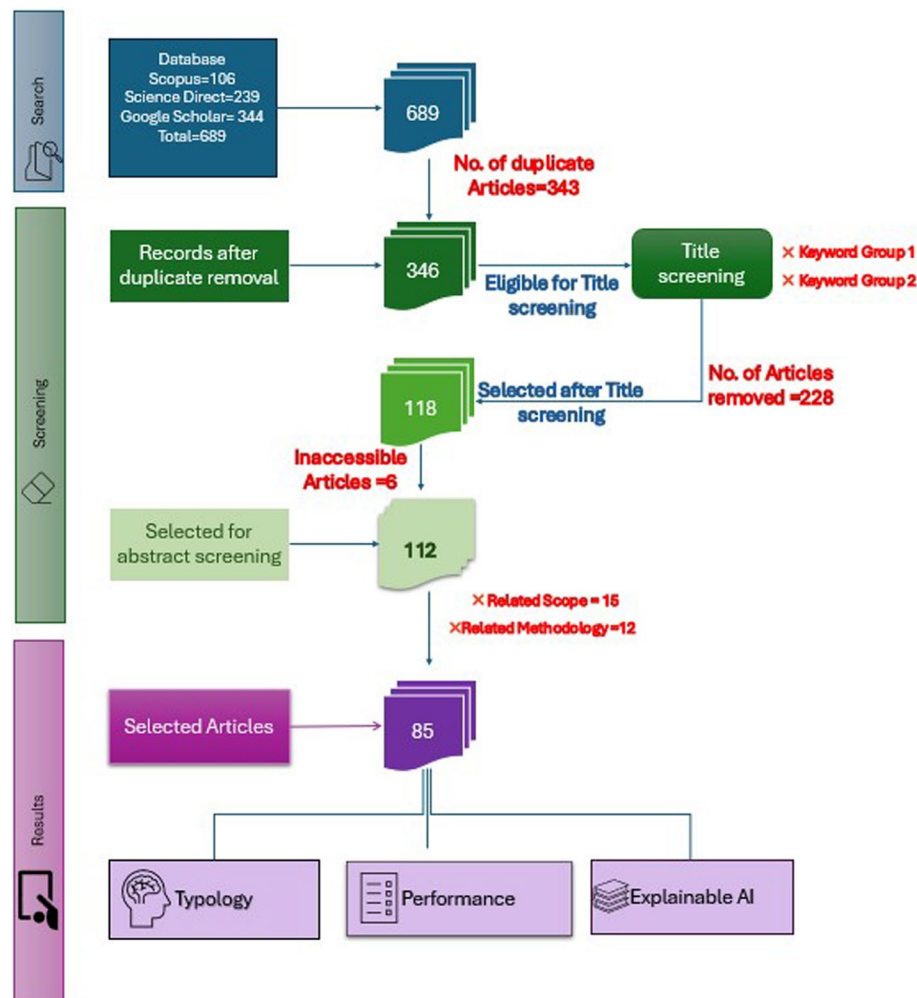


Fig. 1 PRISMA flow diagram

Table 1 Keywords for searching databases

Keyword group	Keywords
Group 1: Urban Heat	Urban heat, Urban heat fluxes, Heat islands, Urban microclimate
Group 2: Application Types	Estimation, Calculations, Assessment, Prediction, Analysis, Quantification, Evaluation, Modelling, Forecasting, Classification
Group 3: Techniques	Machine learning, Deep learning, Data-driven, AI

This review is model-centric and focuses on studies that apply ML and DL models for UHI prediction and classification purposes.

2.2 Search strategy

Figure 1 provides an overview of the various steps involved in this review. The first necessary step was to identify the most relevant concepts to ensure that the review remained focused and addressed specific research questions or objectives [33]. A list of terms from UHI studies was created to search three databases that were deemed to contain the most relevant scientific literature: ScienceDirect, Scopus, and Google Scholar. The search queries combined three groups of keywords, as presented in Table 1, and were run on the titles, abstracts, and keywords of the indexed studies to ensure a comprehensive and

robust search strategy for the literature review. The first keyword group refers to the review theme (for example, UHI and microclimate), the second to the research approach (e.g. estimation, assessment, and modelling), and the third to the techniques or methods used (for example, Machine Learning (ML), Deep Learning (DL), AI, and data-driven). Keywords within a group were always combined with an OR logical operator, whereas keyword groups were always combined using the AND operator. For example:

(Urban heat OR Urban heat fluxes) AND (Estimation OR Calculation) AND (Machine Learning OR Deep Learning)

Different combinations of keywords from each group were generated to acquire more comprehensive results, as instructed by Booth [34]. This SLR considered only studies published between 1 June 2019 and 31 December 2024. The June 2019 cut-off was selected to extend the scope of existing SLRs and systematically review the methodological developments that emerged following the period analysed by Kim and Brown [25].

2.3 Eligibility criteria

Eligible for inclusion in the SLR were peer-reviewed journal articles written in English, published between June 2019 and December 2024; studies that applied ML or DL techniques for UHI classification, prediction, or modelling; studies that focused on presenting details of the model architecture and quantitative evaluation metrics (for example, RMSE, MAE, and R^2); and studies that focused on UHI at any spatial scale (micro, local, meso, and regional). The exclusion criteria were as follows: studies that solely focused on physical models or conventional statistical approaches without ML and DL integration; papers that lacked transparency in methodology, training data, or performance evaluation; and non-peer-reviewed literature, including conference abstracts (if not peer-reviewed), review articles, book chapters, editorials, and theses/dissertations. Moreover, non-English-language publications and studies focusing on heat risk perception, human behaviour, or policy impact without ML or DL modelling were excluded.

2.4 Study selection

The first measure to ensure the relevance of the articles was title screening. At this stage, 28 articles were excluded because they were only marginally or not at all related to the SLR topic. The remaining 118 abstracts were then screened, and any studies that solely relied on physical modelling without combining ML and DL techniques, did not have an ML and DL application as a focal point, or were not linked to the UHI phenomenon were removed. The remaining 112 articles were read in detail and assessed for their eligibility. At this stage, 27 articles were excluded due to insufficient information and equivalent methodology and scope. After all screening steps, 85 studies were included in the analysis to answer the research questions.

3 Characteristics of the paper corpus

Each study was categorised based on the study area, UHI modelling type, horizontal and vertical scale, explainability, error metrics, and ML and DL methods used.

3.1 Spatial distribution of studies

As illustrated in Fig. 2, UHI research is considerably concentrated in Asia, with a significant focus on megacities such as Beijing [18], Seoul [35], and Nanjing [30]. While

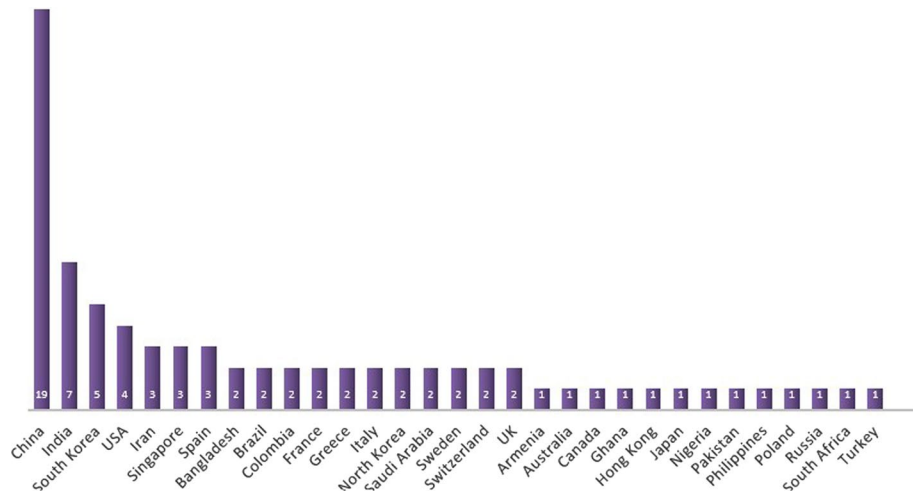


Fig. 2 Spatial distribution of studies

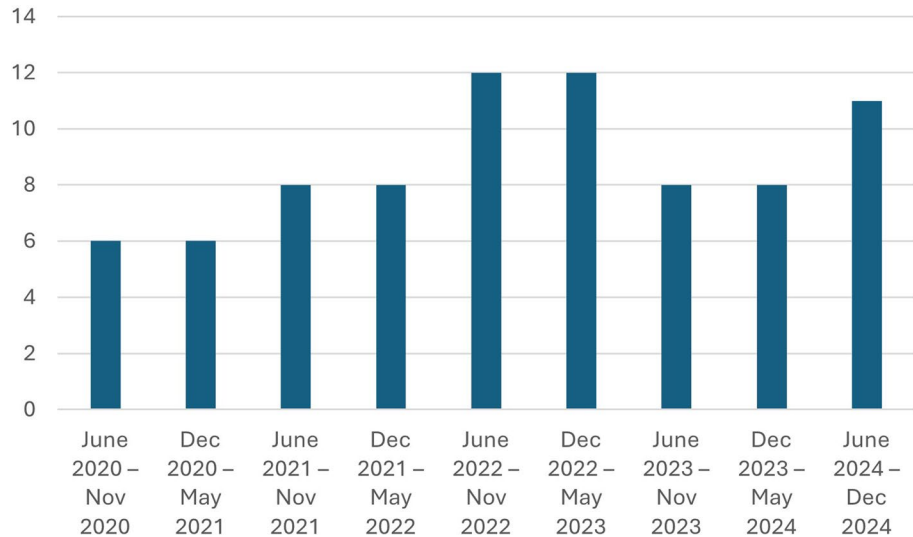


Fig. 3 Year-wise distribution of UHI publications

some studies have been conducted in Europe, considerably less attention has been given to the American and Asian continents, despite increasing concerns over urban heat in cities such as Madrid and Paris. For instance, a recent study examined urban heat in 69 European cities, including major metropolitan centres such as London, Paris, and Madrid [36]. However, there is still scope for further research in this area. This under-representation may indicate a research focus on rapidly urbanising regions, such as Asia, where UHI effects are more pronounced owing to high rates of population growth and urban expansion. For instance, Southeast Asia has been a target region for UHI studies [23]. Conversely, a significant portion of the world, including Africa and South America, which are equally vulnerable to urban heat, remains under-represented in UHI research.

3.2 Year-wise distribution of models in UHI studies

Figure 3 displays the number of UHI-related publications from June 2019 to December 2024. The figure is divided into nine 6-month periods. The graph shows a steady

publishing trend from June 2020 to May 2022, with approximately six to eight publications every six months. There was a sharp increase in publications between June 2022 and May 2023, with 12 publications in each period. This spike might be attributed to the completion and publication of postponed studies because of the pandemic and the heightened global interest in climate change mitigation and urban heat impacts. Factors contributing to the increase in 2022 include advances in remote sensing approaches. Remote sensing (RS) is the process of acquiring information about the surface of the Earth without direct contact [37], such as the use of global satellite data to simulate air temperature over unspecified cities, allowing scalable and computationally effective UHI effect predictions across regions [38]. The publication rate returned to eight publications for the subsequent two periods before rising again to 11 for the last period, reflecting continued interest and research in this field. Finally, the steady rise in the number of publications reflects a sustained and possibly expanding interest in UHI research, likely driven by the growing relevance of urban climate change issues.

4 Insights and discussion

This section interprets the key findings of the systematic review in relation to the three research questions (RQ1–RQ4). It synthesises ML and DL approaches that have been applied in UHI modelling, identifies typological and scale-related trends, and highlights methodological gaps and future opportunities for advancing data-driven urban climate research.

4.1 RQ1-What ML and DL model typologies are used for UHI classification and prediction at different horizontal and vertical scales?

To address this research question, we proceeded in two steps: first, we categorised the methodological approaches used in the selected studies, and then we analysed how these classifications corresponded to different horizontal and vertical scales of the study.

4.1.1 Typologies in ML and DL model design

To systematically assess how ML and DL models contribute to UHI research, we created a typology based on input–output relationships to identify how different model types are structured to capture spatial, temporal, and hybrid dependencies of UHI. This typology provides a structured lens for comparing models and assessing their suitability for different UHI research objectives and uses five groups of models. The groups are named to reflect the type of problem each model is designed to solve and how it handles the data. The names also highlight whether the models combine physical knowledge or use other data types to predict the target. The five main groups are as follows.

1. Spatial Snapshot Models, which focus on static spatial mapping
2. Short-Term Temporal Forecasting Models, which perform immediate forecasting of UHI dynamics
3. Spatio-Temporal Sequence Modelling, which captures multi-step UHI evolution in space and time
4. Physics-Informed Hybrid Models, which integrate ML and DL with physical modelling frameworks and
5. Multi-Modal Multi-Resolution Models, which fuse multiple data sources across scales to model complex urban climate interactions.

This typology is summarised in Table 2, which allows us to analyse how different ML and DL approaches align with the specific demands of the UHI modelling.

4.1.2 Use of model types over time

Figure 4 illustrates the annual use frequency of each model type during the investigated period of this SLR. Early UHI studies (2019–2020) predominantly relied on spatial snapshot models, such as Random Forest and Support Vector Regression, due to their effectiveness in static mapping tasks using remotely sensed indicators e.g., [30, 50, 56]. With the increasing availability of time-series meteorological and in-situ sensor data, researchers began adopting temporal forecasting models to address short-term heat dynamics and diurnal variability. Studies by Cho et al. [35]; Tehrani et al. [36] demonstrate the use of GRU- and LSTM-based architectures for forecasting daily and hourly urban temperatures.

Spatio-temporal sequence models, such as CNN-LSTM and ConvBiLSTM, have gained prominence [28, 95]. Recently, hybrid and multi-modal approaches have emerged to support applications requiring physical consistency, data fusion, or cross-scale

Table 2 Model typology with references

Phase	Model type	Description	References
Spatial Snapshot Models	RF, MLR, SVR, Gradient Boosting, XGboost, K-Mean clustering, etc.	Predicts the target variable based on static spatial features.	Agathangelidis and Cartalis [39]; Ebrahimi and Azadbakht [40]; Athukorala and Murayama [41]; Barthwal and Sharma [42]; Chen et al. [43]; Fashae et al. [44]; Hu et al. [45]; Oh et al. [46]; Qi et al. [47]; Dos Santos [22]; Shafizadeh-Moghadam et al. [48]; Sun et al. [49]; Yu et al. [50]; Alonso and Renard [51]; Adeyemi et al. [52]; Chen et al. [30]; Hidalgo-García and Arco-Díaz [53]; Kafy et al. [54]; Kisters et al. [10]; Erdem Okumus and Terzi [55]; Oliveira et al. [56]; Tepanosyan et al. [57]; Xu et al. [58]; Bansal and Quan [59]; Blunn et al. [60]; Buo et al. [61]; Chang et al. [62]; Fluckiger et al. [63]; Hu et al. [64]; Jato-Espino et al. [65]; Lemoine-Rodríguez et al. [66]; Lyu et al. [67]; Mohammad et al. [68]; Oukawa et al. [69]; Lau and Lin [70]; Li et al. [71]; Gupta and Aithal [72]; Hassani et al. [73]; Koutroumanou-Kontosi et al. [74]; Li et al. [75]; Maithani et al. [76]; Rao et al. [77]; Shao et al. [78]; Shang et al. [79]; Varentsov et al. [80]; Vergara et al. [81]; Bindajam et al. [82]; Ding et al. [83]; Mansourmoghaddam et al. [84]; Oukawa et al. [85]; Zhao et al. [86]; Wang et al. [38]; Du et al. [87]; Fu et al. [19]; Liu et al. [88]; Kavathekar et al. [89]
Temporal Forecasting Models	LSTM, GRU	Forecasts the next-step UHI or temperature using past time series data.	Han et al. [23]; Mendez-Astudillo and Mendez-Astudillo [90]; Huo et al. [91]; Wang et al. [38]
Spatial-Temporal Sequence Models	CNN-LSTM, CONV Bi-LSTM, GANs	Captures spatial and temporal dependencies for forecasting dynamic UHI fields.	Choe and Yom [92]; Cho et al. [93]; Javanroodi et al. [94]; Khan and Maity [95]; Delgado-Enalesa et al. [96]; Li et al. (2023); [75]; Johannsen et al. [97]
Physics-Informed Hybrid Models	WRF + DL, TFT, GANs Hybrid	Weighted fusion of spatial and temporal models with traditional modelling techniques.	Chen et al. [98]; Zhang et al. [99]; Zhu et al. [100]
Multi-Modal and Multi-Resolution Models	Advanced Hybrid Multi-Res	Combines features from multiple data sources (e.g., MODIS + station).	Jia et al. [18]; Wu et al. [101]; Chajaei and Bagheri [102]; Tehrani et al. [36]; Sundar et al. [103]

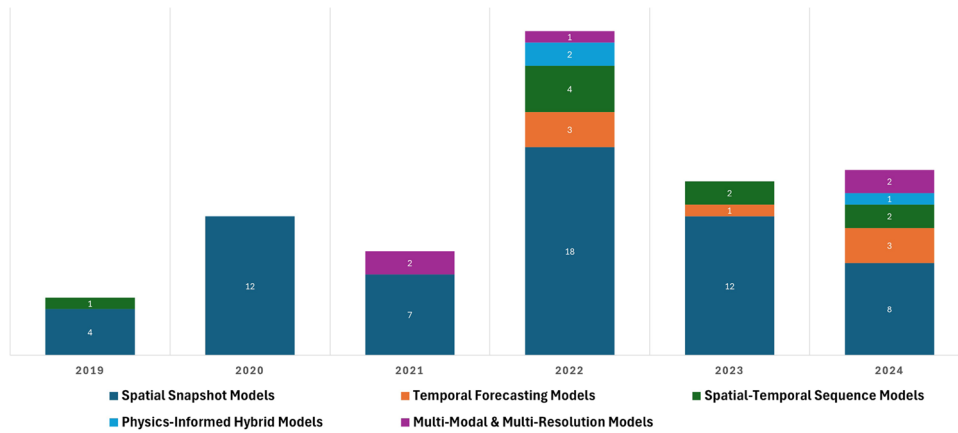


Fig. 4 Annual use of model types during the study period

generalisation, as shown in studies integrating WRF with deep learning [100] or GAN-based data enhancement [61]. Importantly, these model classes coexist rather than replace one another, reflecting a growing methodological toolkit tailored to different UHI modelling needs. While methodological diversity has increased, Fig. 4 also reveals that advanced approaches, such as physics-informed and multi-modal models, remain comparatively underrepresented, indicating persistent challenges in UHI modelling approaches.

4.1.3 Spatial snapshot models

Spatial Snapshot Models are designed to predict UHI patterns at a single point in time based on static spatial features. In this modelling strategy, time is not explicitly considered in the input data; instead, the models generate predictions based on spatial variables, such as land cover, surface characteristics, and urban morphology. These models do not account for changes over time or learn from historical sequences and are therefore limited to providing static "snapshot" views of urban heat patterns. This relationship can be expressed as follows

$$Y_{(\text{long}, \text{lat})} = f(X_{(\text{long}, \text{lat})}) \quad (1)$$

Where Y represents the predicted UHI intensity at a given location (longitude and latitude), and X represents the spatial input features at the same location. Such models are typically used in spatially explicit tasks, such as predicting LST or canopy-level UHI intensity from remotely sensed imagery or urban-form indicators. Commonly used models in this category include RF, multiple linear regression (MLR), and SVR. For example, Chen et al. [30]; Oliveira et al. [56]; Yu et al. [50] used RF and MLR to estimate UHI intensity based on variables such as NDVI, albedo, built-up area, and elevation. While these studies often framed UHI prediction as a regression problem, some employed classification approaches, such as categorising urban areas into high, medium, or low UHI zones. Spatial Snapshot Models offer advantages in terms of computational efficiency and interpretability, making them accessible tools for rapid UHI mapping and policy applications. However, their key limitation is the inability to model temporal dynamics, such as daily heat fluctuations, seasonal cycles, and the progression of

heatwaves. Consequently, although these models remain useful for spatial analysis, they cannot capture the dynamic nature of the UHI.

These models remain prevalent, with the primary objective being spatial pattern mapping rather than temporal prediction. Studies such as those by Yu et al. [50]; Oliveira et al. [56]; Chen et al. [30] demonstrate that these models are primarily selected for spatially explicit UHI mapping tasks, where temporal dynamics are not the primary focus.

4.1.4 Temporal forecasting models

Temporal Forecasting Models aim to predict near-term changes in UHI dynamics by incorporating temporal dimensions into the modelling process. As time-series meteorological and sensor data have become more widely available, UHI modelling has increasingly used these data to move beyond static snapshots and forecast the evolution of urban heat in cities over time. Models in this category typically perform one-step-ahead forecasting, in which the next value in a sequence is predicted based on a window of the historical data. This enables the model to learn temporal dependencies, such as daily or diurnal variations. The predictive relationship is expressed as follows:

$$Y_{(t \rightarrow t+n)} = f(X_{(t-m, t-1)}) \quad (2)$$

where $X_{(t-m, t-1)}$ represents the input features over a past window of time steps, and $Y_{(t \rightarrow t+n)}$ is the predicted UHI intensity n steps in the future. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, are commonly used in this modelling strategy. In addition, hybrid models, such as CNN-LSTM architectures, combine spatial feature extraction with temporal forecasting, enabling more comprehensive representations of UHI dynamics. For example, Cho et al. [35] and Tehrani et al. [36] used GRU-based and ensemble architectures to forecast maximum and minimum daily temperatures. Han et al. [23] proposed a Geo-Kriging-LSTM model to predict pedestrian-level air temperature one hour ahead, integrating historical meteorological data with spatially interpolated environmental factors.

Temporal forecasting models provide important capabilities for capturing the fine-scale temporal dynamics of the UHI, such as diurnal heat fluctuations and short-term temperature variability at the neighbourhood level. However, their ability to model urban-scale complexities, such as interactions across multiple spatial scales and long-term trends, is limited. Their adoption reflects a shift toward applications requiring short-term temporal awareness, such as heat-risk monitoring. Cho et al. [35] and Tehrani et al. [36] show that LSTM and GRU architectures are particularly effective for short-term temperature forecasting, supporting applications such as heat-risk monitoring and early warning systems.

4.1.5 Spatio-temporal sequence models

Spatio-Temporal Sequence Models represent a significant advance over one-step-ahead forecasting approaches, as they are designed to predict multi-step sequences of future UHI values. This capability is important for UHI modelling, which requires maintaining temporal continuity and capturing the evolution of heat patterns [28, 95] that are highly relevant for practical applications, such as short-term heat risk alerts, emergency

response planning, and fine-grained microclimate management. The predictive spatial-temporal relationship can be represented as

$$Y_{(\text{long}, \text{lat}, t)} = f \left(X_{(t-m, t-1)}^{(\text{long}, \text{lat})} \right) \quad (3)$$

Here, the model forecasts the UHI intensity at a specific location and time by learning from a window of historical data at the same location. Beyond the temporal dimension, these models also improve the ability to represent spatial dependencies, such as the influence of the surrounding urban morphology and land use on local UHI conditions.

$$f = f \left(X_{(t-m:t-1)}^{N(x,y)} \right) \quad (4)$$

where $X^N(x, y)$ includes data from neighbouring grid cells or zones around the target location. Architectures such as CNN-LSTM, ConvBiLSTM, temporal GWR, vision transformers, and 3D CNNs are explicitly designed to capture both spatial and temporal dynamics. In these models, the function f incorporates neighbourhood features over a defined temporal window, allowing the model to learn how spatial patterns interact with temporal trends. For example, Han et al. [23] used such an approach to predict temperature at a 1-meter resolution for the next day, combining land use data and weather station observations to capture spatio-temporal UHI dynamics. These models play a crucial role in addressing urban-scale complexities in UHI modelling. They enable finer-grained forecasting that reflects both the spatial heterogeneity of urban areas (e.g. differences between densely built-up zones and green spaces) and the temporal progression of heat phenomena (e.g. diurnal cycles, lagged heat effects, and short-term climatic shifts). As such, Spatio-Temporal Sequence Models offer a strong foundation for multi-dimensional UHI analysis. However, they also face critical limitations: their accuracy depends heavily on the availability of high-resolution continuous spatiotemporal data, which are often uneven across cities. Moreover, although these models excel at capturing local UHI patterns, they are rarely validated across multiple urban contexts, limiting their generalisability and scalability. Addressing this limitation is important for future studies.

These approaches represent an expansion toward modelling the UHI as a dynamic spatial process rather than a static surface. Studies by Khan and Maity [95] and Wang et al. [28] illustrate how CNN-LSTM and BiLSTM models enable multi-step spatio-temporal forecasting, addressing research needs that require both spatial interaction modelling and temporal continuity.

4.1.6 Physics-informed hybrid models

Physics-Informed Hybrid Models aim to improve UHI modelling by combining the strengths of DL with physical modelling and simulations. This strategy is particularly valuable in urban climate research, where physically based models, such as the Weather Research and Forecasting (WRF) model, can capture important atmospheric processes but may lack the flexibility or resolution offered by data-driven ML and DL approaches. The idea is to integrate different model components, each capturing different aspects of the system, into a unified framework. This can be expressed as

$$Y_{(x,y,t)} = w_1 \cdot f_1 \left(S_{(x,y)} \right) + w_2 \cdot f_2 \left(T_t \right) \quad (5)$$

where f_1 represents the physical model output based on spatial inputs ($S(x, y)$), and f_2 is the ML and DL model component that learns from temporal data (T_t). The weights w_1 and w_2 can be learned or manually defined to balance the contributions of each weight. For example, Zhu et al. [100] combined WRF simulations with transformer-based models (TFT) to predict the spatial spread of urban temperatures. This hybrid approach allows the model to learn fine-scale heat patterns while preserving the key physical relationships.

Physics-informed hybrid models are important for addressing the complexities of urban areas. By grounding ML and DL predictions in physical understanding, these models improve generalisability across cities and climates and can better capture processes such as energy balance, urban canopy effects, and vertical heat flux aspects that are difficult to model using data-driven approaches. However, such models are computationally demanding, complex to design, and require careful tuning to ensure their physical consistency. Additionally, their interpretability can be challenging, and their usefulness in operational urban planning is still evolving.

For instance, Zhu et al. [100] demonstrate how integrating WRF outputs with Transformer-based models improves physical consistency and spatial realism in urban temperature prediction.

Although several reviewed studies combine physical models (e.g. WRF) with machine or deep learning architectures, none explicitly implement Physics-Informed Neural Networks (PINNs), where physical governing equations are embedded directly into the training loss function. As such, current UHI research primarily relies on physics-informed hybrid frameworks rather than fully physics-constrained neural models. PINNs represent a promising yet largely unexplored direction for UHI modelling, particularly in terms of enforcing physical consistency and improving generalisability across diverse climatic contexts.

4.1.7 Multi-modal and multi-resolution models

Multi-Modal and Multi-Resolution Models focus on integrating diverse data sources, often at different spatial and temporal scales, to improve the richness and accuracy of UHI modelling. The modelling strategy is as follows:

$$Y_{\text{enhanced}}(t) = \sum w_i \cdot f_i(X_t^{\text{multi-res}}) \quad (6)$$

Here, f_i represents a model component that processes data at a different spatial or temporal resolution, and the weighted sum combines their outputs to produce a more accurate and context-aware prediction of the UHI. This typically involves fusing information from satellite imagery (e.g. MODIS), in situ sensor data, LCZ classifications, land cover maps, and weather station records were used.

Generative Adversarial Networks (GANs) and their variants, such as CycleGANs and DualGANs, are often used in this field. For example, Buo et al. [61] used GANs to generate synthetic temperature data for areas where real data were missing, helping to address data sparsity in urban heat monitoring. Similarly, Sundar et al. [103] and Kong et al. [104] applied CycleGANs and DualGANs to enhance image-based representations and improve urban land classification, which is crucial for downstream UHI modelling tasks.

This modelling strategy is highly relevant for multi-dimensional UHI modelling, as it enables the model to capture interactions across spatial (e.g. neighbourhood-level patterns influenced by city-scale morphology) and temporal scales (e.g. combining long-term trends with high-frequency sensor data). Multi-modal models are particularly important for advancing fine-grained, high-resolution UHI mapping, which is essential for designing targeted heat mitigation interventions.

However, these models face challenges, including the need for careful preprocessing to align heterogeneous data sources, computational intensity, and the tendency to produce outputs that are less transparent to domain experts than traditional models. Moreover, their real-world utility for urban decision-making requires rigorous validation in the field.

The emergence of the Multi-Modal and Multi-Resolution models indicates a growing emphasis on generalisability, physical realism, and data integration. Buo et al. [61] and Sundar et al. [103] exemplify how GAN-based multi-modal frameworks are used to address data sparsity and resolution mismatches by fusing satellite imagery, land-cover data, and in-situ observations.

4.1.8 Horizontal and vertical scales of models

As shown in Table 3, different ML and DL model groups frequently addressed various scales. Spatial Snapshot Models, such as RF, SVM, Gradient Boosting, and Decision Trees, are highly popular across all scales for classification and regression tasks. At the micro and local scales, they are often used to map UHI intensity based on urban morphology and surface features; for example, RF was used by Chen et al. [30]; Oliveira et al. [56]; Yu et al. [50] to predict UHI intensity using NDVI, albedo, and land use indicators. At larger scales, Shao et al. [78] applied XGBoost and Decision Trees for regional and global UHI assessments, demonstrating the scalability of these models for processing large datasets. Short-Term Temporal Forecasting Models, including GRU and LSTM architectures, were used to model short-term UHI dynamics. For example, Cho et al. [35]; Tehrani et al. [36] used GRU-based models to forecast daily maximum and

Table 3 Typology of machine learning models across spatial and functional scales

Typology group	Model	Micro level		Local level		Meso level		Global
		Class.	Pred.	Class.	Pred.	Class.	Pred.	
Spatial snapshot models	Random Forest		X		X		X	X
	Gradient Boosting		X		X		X	
	XGBoost						X	X
	Support Vector Machine	X		X	X	X	X	X
	Decision Tree	X	X	X		X	X	
	Artificial Neural Network	X	X		X	X	X	
Short-term forecasting models	Gated Recurrent Unit							X
Spatio-temporal sequence models	LSTM-BiLSTM		X		X		X	
	Convolutional Neural Network					X		X
Multi-modal/multi-resolution models	GAN		X					
Hybrid spatio-temporal models	Temporal Fusion Transformer							X

minimum temperatures, whereas Han et al. [23] used a Geo-Kriging-LSTM model to predict pedestrian-level air temperature one hour ahead, incorporating spatially interpolated features. These models are used at the meso and local scales, particularly for heat risk monitoring and early warning systems, but are less commonly applied at the micro level because of data unavailability issues.

Spatio-Temporal Sequence Models, such as CNN-LSTM and ConvBiLSTM, with their enhanced capability for capturing UHI evolution across space and time, have been applied at local to meso scales to model dynamic UHI patterns. Han et al. [23] demonstrated this by predicting next-day temperature at 1-meter resolution using CNN-LSTM and ConvBiLSTM architectures. Similarly, Wang et al. [38]; Khan and Maity [95] used such models for meso-scale forecasting, supporting short-term planning and adaptation measures. However, these models were developed using high-resolution spatio-temporal datasets, which limits their generalisation across cities and climates.

Physics-informed hybrid Models, such as WRF combined with DL models, are emerging for regional-scale UHI prediction. Zhu et al. [100] combined WRF with TFT to predict urban temperature fields, showing how physical modelling can be enhanced with data-driven methods to improve generalisability and realism at meso and global scales. These approaches are computationally demanding but offer strong potential for scaling UHI modelling beyond single-city contexts in the future.

Multi-Modal and Multi-Resolution Models, particularly those based on GAN architectures, are gaining traction for enhancing data availability and spatial coverage. Buó et al. [61] used GANs to generate synthetic temperature data in data-scarce urban areas. Sundar et al. [103]; Kong et al. [104] used CycleGANs and DualGANs to enhance urban land classification and imagery quality, contributing to improved inputs for UHI modelling at micro and local scales. However, the use of such models at the meso-or global scale remains limited.

This comparison based on typology shows that while spatial snapshot models remain dominant across scales for static mapping, more advanced DL models, particularly spatio-temporal sequence and hybrid models, are increasingly used to capture UHI dynamics. However, the alignment of model types with urban-scale complexities remains uneven; fine-scale models are often constrained by data availability, whereas cross-scale generalisability remains a key challenge for most DL-based approaches.

Table 4 shows the distribution of ML and DL model types across the vertical layers. For the surface layer, spatial snapshot models such as RF, SVM, ANN, and XGBoost are dominant. These models have been widely used to map static spatial patterns of surface temperature and vegetation indices. For example, Dos Santos (2020) employed RF and SVM to accurately classify surface UHI patterns, while Jia et al. [18] used RF and Gradient Boosting to model surface temperature variations based on land surface and urban morphological indicators. Although these models perform well in spatial mapping tasks, they do not capture the temporal dynamics.

Spatio-Temporal Sequence Models are increasingly applied to the surface layer for prediction tasks. Wang et al. [38] demonstrated that BiLSTM models can effectively model temporal variations in urban air temperatures with high accuracy, providing valuable insights into diurnal UHI dynamics that cannot be captured by static models alone.

For the canopy layer, spatial snapshot models such as RF and XGBoost have been used effectively to model the spatial variability of canopy-air temperatures. Fluckiger et al. [63]

Table 4 ML and DL models used in the classification and prediction of UHI at different vertical levels

Typology Group	Vertical Scale	Task	ML and DL Models Used
Spatial Snapshot Models	Surface Layer	Prediction	Random Forest, Support Vector Machine, Artificial Neural Network, Gradient Boosting, XGBoost
	Surface Layer	Classification	Random Forest, Support Vector Machine, Artificial Neural Network, Decision Tree
	Canopy Layer	Prediction	Random Forest, Artificial Neural Network, XGBoost, Deep Neural Network, Gaussian Process Regression
	Canopy Layer	Classification	Random Forest, Decision Tree
	Boundary Layer	Prediction	Support Vector Machine
Spatio-Temporal Sequence Models	Surface Layer	Prediction	Convolutional Neural Network, Deep Neural Network, BiLSTM
Short-Term Forecasting Models	Canopy Layer	Prediction	Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU)
Multi-Modal and Multi-Resolution Models	Surface Layer	Classification	Generative Adversarial Network

showed that XGBoost outperforms several other models in handling the spatial complexity of canopy-layer temperature fields, especially when integrating environmental and structural variables. Short-term temporal forecasting models are becoming increasingly popular. Tehrani et al. [36] applied GRU and LSTM architectures to model the spatial and temporal variability of canopy-layer air temperatures, showing that these models can capture diurnal heat dynamics and respond to short-term meteorological changes.

GAN-based models, which are multi-modal and multi-resolution, are used in canopy layer studies to enhance classification processes and address data gaps. Ding et al.[83] demonstrated how GANs can be used to generate synthetic temperature data and improve classification accuracy in complex urban canopy environments, particularly when observational data is sparse or unevenly distributed.

In contrast, the modelling of the boundary layer remains limited in current UHI studies. The few examples rely mainly on spatial snapshot Models such as SVM, which have been used for boundary layer temperature prediction. However, the application of more advanced DL models at this scale remains largely unexplored, likely because of the scarcity of high-resolution boundary layer data and the inherent complexity of modelling upper-layer atmospheric processes.

This vertical scale analysis shows that spatial snapshot models are the most widely used across layers, particularly for classification. More advanced DL models, such as spatio-temporal sequence models and short-term temporal forecasting models, are being increasingly applied to the surface and canopy layers, where richer data sources are available. The application of Multi-Modal and hybrid models remains limited but promising, particularly in addressing data gaps at finer vertical scales. Boundary Layer modelling remains a key research gap that future studies should address.

In summary, spatial snapshot models remain dominant across both horizontal and vertical scales, particularly for static UHI mapping. More advanced DL models, such as spatio-temporal sequence models and short-term temporal forecasting models, are being applied more frequently at the surface and canopy layers for dynamic modelling, but are rarely used in the boundary layer. Multi modal and hybrid models are emerging, particularly for addressing data gaps and enhancing physical interactions, but their application remains limited and largely unexplored.

4.2 RQ2-Which computational environments, programming languages, libraries, or software tools were used to implement ML and DL models for UHI classification and prediction?

The SLR revealed a strong preference for Python frameworks, such as Scikit-learn and TensorFlow, which were implemented in 62 studies overview given in Fig. 5. Scikit-learn is a widely used open-source Python library for machine learning that provides simple and efficient tools for data mining, preprocessing, and model evaluation [105]. It is particularly well-suited for implementing traditional ML algorithms, such as classification, regression, and clustering, and is often used in UHI modelling because of its ease of use and integration with scientific computing workflow. TensorFlow is an open-source DL framework developed by Google that is widely adopted for constructing and deploying neural network models for high-dimensional data [106]. It also supports distributed computing, enabling the processing of complex data, such as satellite images and meteorological data, which require high computational resources [107]. Many studies, such as Qi et al. [47]; Khan and Maity [95]; Oliveira et al. [56], rely on Keras, which is a high-level API on top of TensorFlow. It facilitates the development of DL models because of its intuitive interface and pre-built layers for quick experimentation and adaptation to UHI-specific factors. To efficiently process and train models on very large datasets, deep learning architectures such as CNNs, LSTMs, and GANs are typically trained using high-performance computing (HPC) clusters. These clusters provide the computational resources required to handle the intensive processing tasks associated with large-scale data, particularly when using frameworks such as TensorFlow and PyTorch (19 and 12 studies, respectively). For example, Johannsen et al. [97]; Tehrani et al. [36] report using HPC environments to accelerate neural network training workflows that involve high-resolution spatial, temporal, and environmental datasets.

Although Python dominates because of its versatility and extensive libraries for integrating ML models into environmental studies [47], other tools, such as the Google Earth Engine (GEE), are increasingly being used for specialised geospatial processing. GEE provides a cloud-based platform for handling and analysing large-scale satellite and environmental datasets through a distributed infrastructure, making it valuable for urban heat studies that require frequent access to large-volume data. For example, Fashae et al. [44] highlighted the utility of the GEE for processing multi-source geospatial data to support temporal-spatial UHI analysis. Zhao et al. [86] further demonstrated

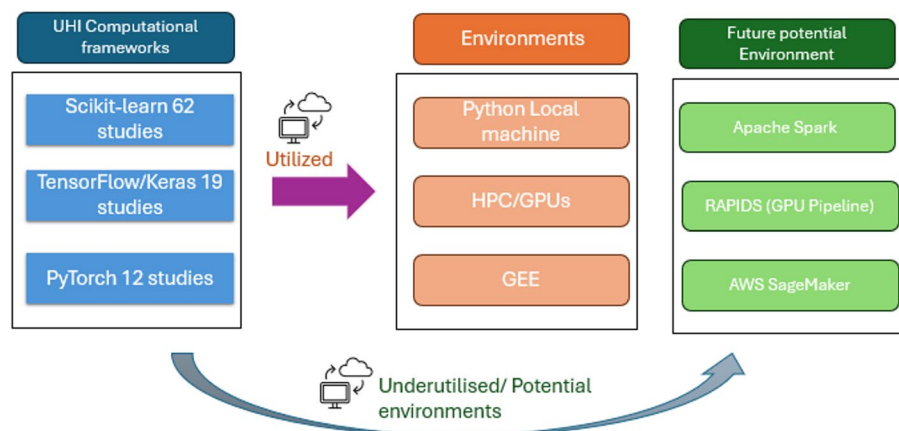


Fig. 5 Computational Environment

that GEE enables urban-scale modelling by bridging local analysis with global datasets via seamless Python integration. Despite these developments, several tools and platforms, such as Dask, Apache Spark, RAPIDS, and hybrid cloud solutions such as AWS SageMaker, remain underutilised in UHI research. These can support distributed processing, improve scalability, and streamline the deployment of real-time or high-frequency modelling processes.

Despite these developments, very few investigations have utilised multi-node distributed systems or hybrid cloud configurations, highlighting the potential for streamlining real-time UHI modelling frameworks in future studies. A gap remains in the general applicability of cloud computing and distributed processing for UHI-related studies, particularly with large datasets that require real-time processing capabilities.

4.3 RQ3-What was the performance of ML and DL models used in UHI studies in terms of error metrics, generalisability, and computational efficiency?

The performance of these models was evaluated using three criteria: error metrics, generalisability, and computational efficiency.

4.3.1 Error metrics

Error metrics are quantitative measures used to assess the accuracy of ML and DL models, ensuring the reliability of the model predictions [108]. The different error metrics adopted in these studies include the root mean square error (RMSE), mean absolute error (MAE), and correlation coefficients. The RMSE calculates the square root of the squared differences between the predicted and observed values and is thus highly sensitive to large errors. The ability to detect large deviations is important in UHI research, where accurate temperature predictions must be obtained to capture heat events [109]. The MAE measures the average absolute deviation of predictions from actual values and is a simple estimate of accuracy that is not subject to a disproportionate influence from outliers [110]. The correlation coefficient quantifies the strength of the relationship between the model predictions and the observed data, which is relevant in UHI studies for determining the dependency among variables from both the urban and environmental factors involved Taylor et al. [111].

RF models have shown consistent and strong performance in UHI studies, with a variation in the RMSE values between 0.27°C and 2°C, and an R^2 of 0.92. This indicates a very strong accuracy in minimising the prediction errors for the diverse environmental variables considered [22, 43, 61, 67]. Recent studies, such as Pande et al. [112], have further refined the capabilities of RF, achieving RMSE values of approximately 1.06°C and an accuracy of approximately 84% for high-resolution temperature predictions, making its application robust for urban air temperatures.

Ensemble Learning techniques, such as Gradient Boosting, XGBoost, and AdaBoost, have been used effectively in predictive UHI modelling. Gradient-boosting models, with RMSEs of approximately 2.5°C and MAE of 1.8°C, have exhibited moderate prediction performance with computational efficiency [22]. Hassani et al. [73] work attests to the application of XGBoost in predictive modelling with achieved RMSE values of approximately 1.15°C and R^2 0.75 for urban heat environment modelling. XGBoost models showed a high prediction efficiency, with an RMSE of 1.15°C and R^2 of 0.75 [73].

AdaBoost, an ML algorithm, exhibited a modest RMSE of 1.50°C and an R^2 of 0.60, making it a strong choice for prediction.

SVM models also exhibited varied performance, with accuracy from 90% to R^2 values from 0.32 to 0.90 Ebrahimi and Azadbakht [40]; Dos Santos [22]. ANN models have good predictive capability, with an RMSE of up to 1.9°C and an MAE of up to 1.4°C Maithani et al. [76]; Mohammad et al. [68]. They can capture intricate UHI patterns and are thus useful for simulating urban areas in UHI models. Likewise, Han et al. [23] employed an ANN to predict microclimate changes in urban areas and obtained an RMSE as low as 1.2°C by including various types of data, such as LULC. DL models, including LSTM, CNNs, and GANs, have demonstrated satisfactory performance in UHI applications. Jia et al. [18]; Khan and Maity [95] say that RNN-based GRU models, being suited for sequence data, yielded consistent temperature prediction outcomes with errors of 1.2°C to 1.8°C (RMSE) and 0.9°C (MAE). In addition, the LSTM models achieved RMSE values of 1.7°C and 1.2°C , respectively. The highest performance was noted in the Bi-LSTM models, with an RMSE as low as 0.11°C , validating their potential for temporal urban temperature prediction. Furthermore, Wang et al. [28] demonstrated the application of LSTM in pedestrian-level temperature prediction, with an RMSE ranging from 0.33°C to 2.08°C . This supports LSTM's potential of LSTMs in high-resolution UHI modelling. CNNs have also been used in UHI studies, with RMSE ranging from 1.1°C to 2°C . CNNs exhibited excellent classification performance with precision scores of 0.86 and AUC-ROC of 0.88; thus, they are suitable for detecting spatial temperature differences in urban areas.

GANs, particularly CycleGAN combined with SVM, yielded excellent predictive accuracy (0.98) and AUC-ROC of 0.99; thus, it is one of the best methods for simulating urban climates at an advanced level.

The error statistics show the accuracy of each method; however, comparing them is difficult because these models are applied under different conditions. This highlights the need for standard datasets and benchmarks in UHI research. In the future, efforts should focus on creating comparative frameworks that account for the unique conditions of each study population.

Moreover, improving reproducibility is essential for making these comparisons meaningful. Many studies report overall accuracy values but provide limited details on data preprocessing, validation schemes and model parameters. Without this information, it is difficult for other researchers to replicate the findings or assess the model robustness under different conditions. Therefore, future research should promote transparent and reproducible evaluation pipelines, where datasets, error metrics, and hyperparameters are clearly documented and openly shared. Such practices would support cumulative scientific progress and help transform UHI modelling into a more collaborative and verifiable research area.

4.3.2 Generalisability

Generalisability refers to the ability of a model to perform reliably across different urban environments, climatic conditions, spatial scales, and city sizes, thereby demonstrating its robustness to new data or changing conditions in UHI applications [29, 67]. The review shows that most UHI studies focus on local, single-city case applications and rarely evaluate whether the trained models remain valid when applied to other cities,

regions, or climatic settings. For example, studies by Chen et al. [30]; Alonso and Renard [51]; Xu et al. [58] applied their models to specific regions without testing or claiming applicability beyond the study area.

A small number of studies, such as Erdem Okumus and Terzi [55]; Fashae et al. [44] and Maithani et al. [76], discuss the potential transferability conceptually but do not empirically evaluate model performance across diverse urban contexts, leaving generalisability largely unverified. Only a limited subset of studies, such as [91, 92], examine regional applicability, and even these do not implement explicit transfer learning frameworks or systematic cross-city validation. While Zhao et al. [86] consider generalisability at the design level, empirical testing across contrasting urban and climatic settings remains absent. Overall, the results indicate that cross-city transferability and formal transfer learning remain largely unexplored, representing a critical gap in current ML- and DL-based UHI modelling.

4.3.3 Computational efficiency

Computational efficiency focuses on the processing demands of ML and DL models, indicating how quickly and resource-effectively a model can process the large datasets required for UHI modelling [91, 93]. This aspect is essential for determining the feasibility of the model for real-time and large-scale applications. Most of the studies we reviewed did not explain how well their models handled large amounts of data. In many cases, researchers use their local computers without discussing whether their methods can work efficiently for larger projects or in real time [91, 93]. Some mentioned faster performance when using advanced techniques such as deep learning or ensemble methods [101], but very few discussed the use of cloud platforms or shared computing systems. For example, Lyu et al. [67] used strong local computers to run complex models but did not explore how well this setup would scale up to larger datasets or different cities.

However, there are only a few examples of this. Fashae et al. [44]; Rao et al. [77]; Shang et al. [79] used GEE to process large satellite and environmental datasets more quickly, showing the benefits of cloud-based platforms. Ding et al. [83] used smart statistical techniques to handle weather data efficiently, suggesting their approach could work well for larger-scale applications too. Sundar et al. [103] ran a hybrid model on Google Colab and showed that using cloud tools can help make model training faster and more flexible. In general, most studies have not focused on computational efficiency. If ML and DL models are to be useful for real-world urban heat planning, future research should focus on how to make these models faster, more efficient, and ready to work with larger and more complex datasets.

4.4 RQ4-What explainability techniques have been used in ML and DL-based UHI modelling?

Many recent DL models, such as Temporal Fusion Transformers (TFTs) and GANs, achieve high predictive accuracy but often function as black boxes, making it difficult to interpret how predictions are derived. This lack of transparency limits their usefulness for planners and policymakers who need to understand why specific areas emerge as heat hotspots in the first place.

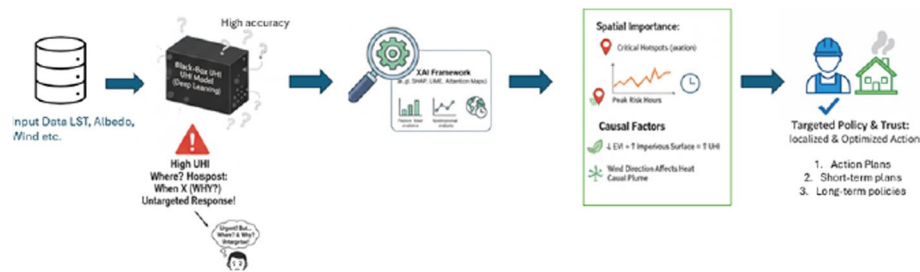


Fig. 6 Importance of XAI

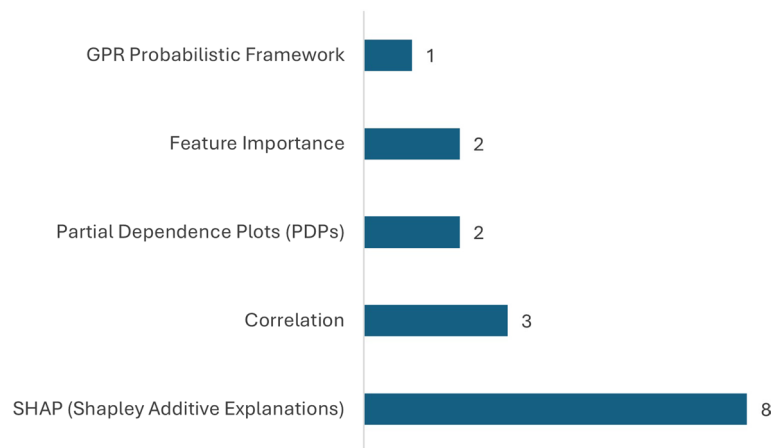


Fig. 7 Techniques used to explain ML and DL model results

XAI methods address this challenge by identifying the location, timing, and reasons for the occurrence of extreme heat. Techniques such as SHAP, LIME, attention maps, or Grad-CAM quantify variable importance and reveal the spatial and temporal drivers behind model outputs. As illustrated in Fig. 6, these methods convert an opaque predictive model into a practical decision-support tool, enabling targeted, trustworthy, and actionable heat mitigation strategies (Fig. 7).

This section examines the use of explainability techniques in UHI modelling to make model outputs more obvious, transparent, and relevant for real-world judgments.

Interpretability and explainability are closely related but distinct concepts in machine learning and deep learning. Interpretability refers to the degree to which a human can understand the internal mechanics of a model, often achievable with simpler models such as linear regression or decision trees, where the input–output relationships are straightforward [113]. In contrast, explainability focuses on understanding the decisions made by complex “black-box” models, whose internal workings are not easily accessible. In such cases, techniques such as SHAP, LIME, and Grad-CAM are employed to provide post-hoc explanations that clarify the model outputs. These methods are especially critical in UHI studies, where transparency is essential for building trust in the model predictions and ensuring their practical usability. Although various techniques exist to address this challenge, much of the current research relies on opaque models.

16 out of 85 studies adopted techniques to explain their outputs, as shown in Fig. 7. Ongoing efforts are required to enhance both the interpretability of the model design and the explainability of the model results [114]. In conclusion, this review shows that

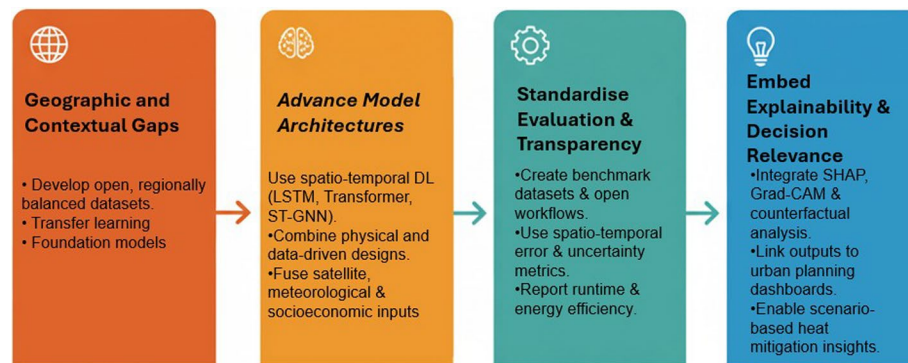


Fig. 8 Future research directions

many UHI studies rely on inherently interpretable models, such as RF and MLR, and the increasing use of DL architectures, such as LSTM, CNNs, GANs, and Transformers, requires more robust explainability methods. However, the use of XAI in UHI research remains restricted and primarily retroactive, with methods such as SHAP and attention weights applied post-hoc rather than being inherent to the design of the models. Crucially, most existing techniques do not provide spatio-temporal explainability, which means they do not indicate where and when specific factors influence UHI forecasts. This is a significant issue for urban planners who want not only accurate models but also clear models that explain why a hotspot is projected in a specific location at a given moment. Without this level of explainability, it is difficult to connect model outputs to concrete measures such as targeted greening, zoning choices, or emergency response plans. Future UHI models must include spatiotemporal explainability to enable educated, localised, and time-sensitive decision-making for urban climate adaptation and heat-resilience planning.

5 Gaps and future research directions

Building on the findings presented in Chapters 3 and 4, this chapter synthesises the main research gaps and proposes future directions for advancing ML- and DL-based UHI modelling. Each subsection explicitly responds to the gaps mentioned in the spatial distribution of studies and three Research Questions (RQs) and translates empirical findings into actionable pathways for future studies. Figure 8 summarises these directions.

5.1 Geographic and contextual biases

As shown in Fig. 2, the geographic distribution of ML- and DL-based UHI studies is highly uneven, with a strong concentration in China and the United States, followed by a smaller number of studies in Europe and East Asia. Only a limited number of investigations have examined cities in Latin America (Brazil, $n = 2$; Colombia, $n = 1$), Africa (Nigeria, Ghana, and South Africa, each $n = 1$), and Eastern Europe (Poland, $n = 1$; Estonia, $n = 1$). Most studies have focused on large, data-rich metropolitan areas, such as Beijing, Seoul, Tokyo, Mumbai, Tehran, and London, where dense meteorological networks and long-term remote sensing archives are available.

This geographic concentration highlights a contextual imbalance in current UHI research, with limited empirical evidence from tropical, arid, and data-scarce urban environments. Consequently, ML and DL models trained and validated primarily in

temperate, data-rich cities may exhibit reduced predictive performance and limited generalisability when applied to cities with different climatic regimes, urban morphologies, and data constraints. Consequently, cities that are often most vulnerable to extreme heat remain underrepresented, and the model performance in these contexts remains largely untested.

5.2 Methodological and architectural gaps

As discussed in Sect. 4.1.1, UHI models vary widely in design, yet many studies lack transparent reporting of their architecture or assumptions. However, three key methodological issues persist.

5.2.1 Dominance of static regression frameworks

Approximately 45% of studies rely on regression-based or other static ML models that cannot capture the diurnal and seasonal variations. These models overlook lag effects, anthropogenic heat, and feedback mechanisms. Future work should employ spatio-temporal architectures, such as LSTMs, GRUs, ConvLSTMs, Transformers, or ST-GNNs, that learn dependencies across time and space. Attention mechanisms can highlight critical past events, whereas hybrid CNN-RNN networks can capture both spatial and temporal features. Scenario examples (with evaluation goals):

- Heatwave-sequence prediction: use an encoder–decoder LSTM to forecast 72-hour heatwave intensity from 10 days of prior data; evaluate across diurnal phases to test temporal generalisation.
- Multi-city forecasting: apply a Transformer to predict daily heat across several cities and measure accuracy decay with distance or climatic dissimilarity. These scenarios allow researchers to identify when and where models fail, a limitation highlighted in Sect. 4.3.1, and to guide the development of more transferable architectures.

5.2.2 Under-use of hybrid and physics-informed architectures

Only 9 of 85 studies combine physically based and data-driven models. Physically based models (e.g. WRF and ENVI-met) embed process knowledge but struggle with incomplete data, whereas DL models learn complex patterns but may ignore physical constraints. Develop hybrid and physics-informed models that merge simulation outputs (e.g. WRF heat flux fields) using DL architectures. PINNs should encode the energy balance or diffusion equations directly into the loss function to maintain physical realism. Comparative benchmarking against pure DL models must be performed to verify whether the integration of physics improves the stability under unseen climatic conditions.

5.2.3 Limited integration of diverse inputs

About 73% of studies rely on a single input type (e.g., satellite LST or air-temperature indices), limiting their ability to capture multi-factor interactions. Only 18% use advanced models capable of handling multi-source data. Advanced multimodal fusion models (Transformers, Informer/Autoformer, ST-GNNs, or PINNs) can combine satellite, meteorological, socioeconomic, and land-use data. Ablation studies should quantify the contribution of each data source to the overall predictive gain, ensuring model interpretability and efficient data prioritisation.

5.3 Performance, metrics, and computational transparency

5.3.1 Evaluation metrics, benchmarking, and uncertainty

As shown in Sect. 4.3.1, approximately 81% of studies rely on global RMSE or MAE scores, masking spatial and temporal weaknesses. Future evaluations should incorporate spatio-temporal error diagnostics and explicitly report uncertainty.

- Time-resolved metrics (hourly or seasonal error curves) to detect recurring biases.
- Geographically weighted RMSE or Local MAE to reveal land-use-specific errors.
- Spatial error heatmaps to visualise accuracy across urban zones.
- Confidence intervals or predictive-uncertainty maps to quantify reliability. Moreover, future studies should establish benchmark datasets and baseline models to enable consistent cross-study comparisons and performance rankings.

5.3.2 Generalisability and cross-city validation gaps

This review found little empirical evidence that current ML- and DL-based UHI models generalise reliably across cities. Most studies train and evaluate models within single-city case studies. Although some studies use multi-city datasets, they typically rely on pooled training rather than explicit cross-city validation.

Consequently, the performance of models across different urban forms, climatic conditions, and data regimes remains largely untested. This lack of systematic cross-city evaluation limits the applicability of ML and DL models to data-rich and well-instrumented cities.

Limited cross-city validation represents a key gap in current UHI modelling research and constrains the operational deployment of ML- and DL-based approaches. Addressing this gap requires future work to prioritise standardised cross-city benchmarks and evaluation frameworks.

5.3.3 Scalable and transparent computational environments

Most reviewed studies (see Sect. 4.3.3) lack details on training time, hardware, and software dependencies, often relying on offline or local environments that hinder their scalability. Future studies should adopt open, cloud-based, and reproducible workflows, and computational platforms should be documented with reproducible environment files and version-tracking. Using pretrained geospatial models (e.g. Prithvi and SatMAE) can substantially reduce training costs and support the transfer to data-sparse cities in the future. Each study should also report runtime benchmarks (training time per epoch, energy consumption and GPU hours) to ensure environmental and computational transparency. This is directly connected to the need for scalability metrics, as discussed in Sect. 4.2.

5.4 Explainability, counterfactuals, and decision support

As noted in Sect. 4.4, approximately 46% of DL studies (LSTMs, CNNs, Transformers, GANs) function as “black boxes”, limiting interpretability. This undermines the confidence of planners who require justifiable insights into the future. Explainability should be embedded into model pipelines rather than being added post-hoc. Future research should integrate the following:

- Feature attribution (SHAP, LIME) to quantify variable importance [114, 115].

- Spatial explanations (Grad-CAM, Grad-CAM++) to visualise influential map regions.
- Surrogate models (decision trees) to approximate complex model logic [116].
- Counterfactual analysis to explore causal “what-if” scenarios [117].

5.5 Practical implications for urban planning and decision-making

The reviewed ML and DL approaches demonstrate clear potential to support urban planning by generating high-resolution heat maps, identifying the dominant drivers of urban heat (e.g. land cover and urban form), and evaluating mitigation strategies such as greening or building configuration changes [49, 75, 118]. These outputs are well-suited for identifying heat hotspots and prioritising areas for climate adaptation.

However, the review shows that most ML- and DL-based UHI models remain primarily used in academic or experimental contexts rather than being routinely embedded in planning or policy workflow. While many studies claim planning relevance, they often stop at producing maps, indicators, or simulations without demonstrating how these outputs inform zoning regulations, design standards, or official decision-making processes [59, 83]. Even widely adopted frameworks, such as Local Climate Zones and WUDAPT, are more frequently used to illustrate their potential usefulness than to support their operational implementation by city authorities (WUDAPT Collaboration, 2020).

A limited number of studies have moved closer to practice by supporting climate adaptation planning or early-stage design evaluation through fast or surrogate modelling approaches [102]. Nevertheless, these applications largely remain at the prototype level, rely on expert interpretation, and show little evidence of sustained or routine use by municipalities.

The main practical challenge identified by this review is not the lack of advanced ML or DL methodologies but the weak integration between research outputs and planning practices. Improving the real-world impact will require closer collaboration between researchers and planners, clearer alignment of model outputs with planning-relevant indicators, and integration into operational tools, such as urban digital twin platforms. Without such integration, ML- and DL-based UHI models are likely to remain scientifically robust but underutilised in real-world urban decision-making.

6 Conclusion

This systematic literature review analysed the development of ML and DL approaches for UHI modelling from mid-2019 to the end of 2024. This review shows that data-driven methods have substantially expanded the analytical capabilities of UHI studies, particularly in terms of spatial detail, short-term forecasting, and the integration of heterogeneous urban and remote-sensing data. Simultaneously, our findings reveal that progress has been uneven across modelling objectives, spatial–temporal scales, and levels of methodological maturity.

From a modelling perspective, static spatial snapshot approaches continue to dominate UHI mapping and classification tasks despite increasing data availability and the recognised importance of temporal dynamics. More recent spatio-temporal and hybrid DL architectures have begun to address these limitations by incorporating temporal dependencies and multi-scale interactions. However, their application remains largely

restricted to data-rich, single-city case studies, which limits the generalisability of current models across different urban forms and climatic contexts. This review further highlights the persistent shortcomings in evaluation and interpretability. Most studies rely on global performance metrics without analysing spatial or temporal error patterns, which constrains the understanding of model reliability under varying conditions. In addition, the limited adoption of explainable and physics-informed approaches restricts the transparency of complex DL models, reducing their suitability for urban planning and policy-oriented applications, where interpretability is essential.

The findings indicate that future progress in ML- and DL-based UHI modelling depends not only on improving predictive accuracy, but also on developing models that can be trained and transferred across multiple cities, evaluated using spatial–temporal error diagnostics rather than global accuracy metrics alone, and interpreted through embedded explainability or physics-informed constraints that link model behaviour to urban processes. Strengthening cross-city validation, embedding physical constraints, and adopting reproducible workflows are critical steps toward developing robust and decision-ready UHI models. By synthesising current practices and clearly identifying unresolved challenges, this review provides a focused foundation for guiding future research toward ML and DL frameworks that are both scientifically robust and operationally relevant for urban heat mitigation and climate-resilient city planning.

Author contributions

N.S. designed the study, conducted the systematic literature review, and wrote the initial draft of the manuscript. M.F. and F.O. provided methodological guidance and contributed to conceptual refinement and the interpretation of results. R.Z.M. supervised the overall research design and provided critical revisions to improve clarity and coherence. All authors reviewed and approved the final version of the manuscript.

Funding

This research received no external funding.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate:

Not applicable.

Consent for publication:

Not applicable

Clinical trial number:

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 12 November 2025 / Accepted: 13 March 2026

Published online: 27 March 2026

References

1. Weng Q, Lu D, Schubring J. Estimation of land surface temperature-vegetation abundance relationship for urban heat island studies. *Remote Sens Environ.* 2004;89(4):467–83. <https://doi.org/10.1016/j.rse.2003.11.005>.
2. Santamouris M, Cartalis C, Synnefa A, Kolokotsa D. On the impact of urban heat island and global warming on the power demand and electricity consumption of buildings-a review. *Energy and Buildings.* 2015;98:119–24. <https://doi.org/10.1016/j.enbuild.2014.09.052>.
3. World Health Organization. *Ambient Air Pollution: A Global Assessment of Exposure and Burden of Disease.* Geneva, Switzerland: World Health Organization; 2018.
4. Li X-X, Norford LK. Evaluation of cool roof and vegetations in mitigating urban heat island in a tropical city, singapore. *Urban Climate.* 2016;16:59–74. <https://doi.org/10.1016/j.uclim.2015.12.002>.
5. Chaturvedi V, Vries WT. Machine learning algorithms for urban land use planning: A review. *Urban Science.* 2021;5(3):3. <https://doi.org/10.3390/urbansci5030068>.

6. Oke TR. City size and the urban heat island. *Atmos Environ* (1967). 1973;7(8):769–79. [https://doi.org/10.1016/0004-6981\(73\)90140-6](https://doi.org/10.1016/0004-6981(73)90140-6).
7. Oke TR. Towards better scientific communication in urban climate. *Theor Appl Climatol*. 2006;84(1):179–90. <https://doi.org/10.1007/s00704-005-0153-0>.
8. Weng Q. Thermal infrared remote sensing for urban climate and environmental studies: methods, applications, and trends. *ISPRS J Photogramm Remote Sens*. 2009;64(4):335–44. <https://doi.org/10.1016/j.isprsjprs.2009.03.007>.
9. Gartland LM. *Heat Islands: Understanding and Mitigating Heat in Urban Areas*. London: Routledge; 2008. <https://doi.org/10.4324/9781849771559>.
10. Kisters P, Ngu V, Edinger J. Urban heat island detection utilizing citizen science. *Advances in Service-Oriented and Cloud Computing*. 2022;1617:94–8. <https://doi.org/10.1007/978-3-031-23298-5>.
11. Janiesch C, Zschech P, Heinrich K. Machine learning and deep learning. *Electron Mark*. 2021;31(3):685–95. <https://doi.org/10.1007/s12525-021-00475-2>.
12. Sarker IH. Deep learning: A comprehensive overview on techniques, taxonomy, applications and research directions. *SN Computer Science*. 2021;2(6):420. <https://doi.org/10.1007/s42979-021-00815-1>.
13. Alzubaidi L, Zhang J, Humaidi AJ, Al-Dujaili A, Duan Y, Al-Shamma O, et al. Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *J Big Data*. 2021;8(1):53. <https://doi.org/10.1186/s40537-021-00444-8>.
14. Jordan MJ, Mitchell TM. Machine learning: trends, perspectives, and prospects. *Science*. 2015;349(6245):255–60. <https://doi.org/10.1126/science.aaa8415>.
15. Sahana V, Mondal A, Sreekumar P. Drought vulnerability and risk assessment in india: Sensitivity analysis and comparison of aggregation techniques. *J Environ Manage*. 2021;299:113689. <https://doi.org/10.1016/j.jenvman.2021.113689>.
16. Zhuang F, Qi Z, Duan K, Xi D, Zhu Y, Zhu H, et al. A comprehensive survey on transfer learning. *Proc IEEE*. 2021;109(1):43–76. <https://doi.org/10.1109/JPROC.2020.3004555>.
17. Shi Y, Ren C, Luo M, Ching J, Li X, Bilal M, et al. Utilizing world urban database and access portal tools (wudapt) and machine learning to facilitate spatial estimation of heatwave patterns. *Urban Climate*. 2021;36:100797. <https://doi.org/10.1016/j.uclim.2021.100797>.
18. Jia H, Yang D, Deng W, Wei Q, Jiang W. Predicting land surface temperature with geographically weighed regression and deep learning. *WIREs Data Min Knowl Discovery*. 2021;11(1):1396. <https://doi.org/10.1002/widm.1396>.
19. Fu S, Wang L, Khalil U, Cheema AH, Ullah I, Aslam B, et al. Away forward for a sustainable environment. *Physics and Chemistry of the Earth Parts A/B/C*. 2024;135:103682. <https://doi.org/10.1016/j.pce.2024.103682>.
20. Alhawiti RH. Using landsat-8 data to explore the correlation between urban heat island and urban land uses. *Int J Res Eng Technol*. 2016;05(03):457–66. <https://doi.org/10.15623/ijret.2016.050308>.
21. Chakraborty T, Sarangi C, Lee X. Reduction in human activity can enhance the urban heat island: insights from the covid-19 lockdown. *Environ Res Lett*. 2021;16(5):054060. <https://doi.org/10.1088/1748-9326/abef8e>.
22. Dos Santos RS. Estimating spatio-temporal air temperature in london (uk) using machine learning and earth observation satellite data. *Int J Appl Earth Obs Geoinf*. 2020;88:102066. <https://doi.org/10.1016/j.jag.2020.102066>.
23. Han J, Chong A, Lim J, Ramasamy S, Wong NH, Biljecki F. Microclimate spatio-temporal prediction using deep learning and land use data. *Build Environ*. 2024;253:111358. <https://doi.org/10.1016/j.buildenv.2024.111358>.
24. Deilami K, Kamruzzaman M, Liu Y. Urban heat island effect: A systematic review of spatio-temporal factors, data, methods, and mitigation measures. *Int J Appl Earth Obs Geoinf*. 2018;67:30–42. <https://doi.org/10.1016/j.jag.2017.12.009>.
25. Kim SW, Brown RD. Urban heat island (uhi) intensity and magnitude estimations: a systematic literature review. *Sci Total Environ*. 2021;779:146389. <https://doi.org/10.1016/j.scitotenv.2021.146389>.
26. Wu Y, Mashhoodi B, Patuano A, Lenzholzer S, Narvaez Zertuche L, Acred A. Heat-prone neighbourhood typologies of european cities with temperate climate. *Sustain Cities Soc*. 2022;87:104174. <https://doi.org/10.1016/j.scs.2022.104174>.
27. Zhou H, Ma Y, Qin J, Song W, Ren C. Urban heat mitigation strategies for different local climate zones: A comparative simulation study in three chinese cities. *Energy and Buildings*. 2023;290:113353. <https://doi.org/10.1016/j.enbuild.2023.113353>.
28. Wang H, Zhang J, Yang J. Time series forecasting of pedestrian-level urban air temperature by lstm: Guidance for practitioners. *Urban Climate*. 2024;56:102063. <https://doi.org/10.1016/j.uclim.2024.102063>.
29. Tang J, Lan X, Lian Y, Zhao F, Li T. Estimation of urban–rural land surface temperature difference at different elevations in the Qinling–Daba mountains using MODIS and the random forest model. *Int J Environ Res Public Health*. 2022;19(18):11442. <https://doi.org/10.3390/ijerph191811442>.
30. Chen S, Yang Y, Deng F, Zhang Y, Liu D, Liu C, et al. A high-resolution monitoring approach of canopy urban heat island using a random forest model and multi-platform observations. *Atmos Meas Tech*. 2022;15(3):735–56. <https://doi.org/10.5194/amt-15-735-2022>.
31. Kapucu N, Ge Y, Rott E, Isgandar H. Urban resilience: multidimensional perspectives, challenges and prospects for future research. *Urban Gov*. 2024;4(3):162–79. <https://doi.org/10.1016/j.ugj.2024.09.003>.
32. Longo L, Brcic M, Cabitza F, Choi J, Confalonieri R, Del Ser J, et al. Explainable artificial intelligence (xai) 2.0: A manifesto of open challenges and interdisciplinary research directions. *Inf Fusion*. 2024;106:102301. <https://doi.org/10.1016/j.inffus.2024.102301>.
33. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The prisma 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*. 2021;372:71. <https://doi.org/10.1136/bmj.n71>.
34. Booth A. Searching for qualitative research for inclusion in systematic reviews: A structured methodological review. *Syst Rev*. 2016;5(1):74. <https://doi.org/10.1186/s13643-016-0249-x>.
35. Cho D, Yoo C, Im J, Cha D. Comparative assessment of various machine learning-based bias correction methods for numerical weather prediction model forecasts of extreme air temperatures in urban areas. *Earth Space Sci*. 2020;7(4):2019–000740. <https://doi.org/10.1029/2019EA000740>.
36. Tehrani AA, Veisi O, Delavar Y, Bahrami S, Sobhaninia S, Mehan A, et al. Predicting urban heat island in European cities: A comparative study of gru, dnn, and ann models using urban morphological variables. *Urban Climate*. 2024;56:102061.
37. Lillesand T, Kiefer RW, Chipman J. *Remote sensing and image interpretation*. 7th ed. Hoboken, NJ, USA: John Wiley & Sons; 2015.

38. Wang S-Y, Ou H-Y, Chen P-C, Lin T-P. Implementing policies to mitigate urban heat islands: Analyzing urban development factors with an innovative machine learning approach. *Urban Climate*. 2024;55:101868. <https://doi.org/10.1016/j.uclim.2024.101868>.
39. Agathangelidis I, Cartalis C. Improving the disaggregation of modis land surface temperatures in an urban environment: a statistical downscaling approach using high-resolution emissivity. *Int J Remote Sens*. 2019;40(13):5261–86. <https://doi.org/10.1080/01431161.2019.1579386>.
40. Ebrahimi H, Azadbakht M. Downscaling modis land surface temperature over a heterogeneous area: An investigation of machine learning techniques, feature selection, and impacts of mixed pixels. *Computers & Geosciences*. 2019;124:93–102. <https://doi.org/10.1016/j.cageo.2019.01.004>.
41. Athukorala D, Murayama Y. Spatial variation of land use/cover composition and impact on surface urban heat island in a tropical sub-saharan city of Accra, Ghana. *Sustainability*. 2020;12(19):7953. <https://doi.org/10.3390/su12197953>.
42. Barthwal A, Sharma A. Analysis and prediction of urban ambient and surface temperatures using machine learning approach. *Int J Syst Assur Eng Manag*. 2022;13(1):1–10. <https://doi.org/10.1007/s13198-021-01502-3>.
43. Chen S, Zhang W, Wong NH, Ignatius M. Combining citygml files and data-driven models for microclimate simulations in a tropical city. *Build Environ*. 2020;185:107314. <https://doi.org/10.1016/j.buildenv.2020.107314>.
44. Fashae OA, Adagbasa EG, Olusola AO, Obateru RO. Land use/land cover change and land surface temperature of ibadan and environs, nigeria. *Environ Monit Assess*. 2020;192(2):109. <https://doi.org/10.1007/s10661-019-8054-3>.
45. Hu D, Meng Q, Zhang L, Zhang Y. Spatial quantitative analysis of the potential driving factors of land surface temperature in different “centers” of polycentric cities: a case study in Tianjin, china. *Sci Total Environ*. 2019;706:135244. <https://doi.org/10.1016/j.scitotenv.2019.135244>.
46. Oh JW, Ngarambe J, Duhirwe PN, Yun GY, Santamouris M. Using deep-learning to forecast the magnitude and characteristics of urban heat island in seoul korea. *Sci Rep*. 2020;10(1):3559. <https://doi.org/10.1038/s41598-020-60632-z>.
47. Qi P, Cui Y, Zhang H, Hu S, Yao L, Li LB. Evaluating multivariable statistical methods for downscaling nighttime land surface temperature in urban areas using *ml*. *IEEE Access*. 2020;8:162085–98. <https://doi.org/10.1109/ACCESS.2020.3021034>.
48. Shafizadeh-Moghadam H, Weng Q, Liu H, Valavi R. Modeling the spatial variation of urban land surface temperature (*lst*) in relation to environmental and anthropogenic factors: A case study of tehran, iran. *GIScience & Remote Sensing*. 2020;57(4):483–96. <https://doi.org/10.1080/15481603.2020.1736857>.
49. Sun F, Liu M, Wang Y, Wang H, Che Y. The effects of *3d* architectural patterns on the urban surface temperature at a neighborhood scale: Relative contributions and marginal effects. *J Clean Prod*. 2020;258:120706. <https://doi.org/10.1016/j.jclepr.2020.120706>.
50. Yu Z, Chen S, Wong NH, Ignatius M, Deng J, He Y, et al. Dependence between urban morphology and outdoor air temperature: A tropical campus study using random forests algorithm. *Sustain Cities Soc*. 2020;61:102200. <https://doi.org/10.1016/j.scs.2020.102200>.
51. Alonso L, Renard F. A new approach for understanding urban microclimate by integrating complementary predictors at different scales in regression and machine learning models. *Remote Sens*. 2020;12(15):2434. <https://doi.org/10.3390/rs12152434>.
52. Adeyemi A, Ramoelo A, Cho MA, Strydom J. Assessment of spatio-temporal direction of impervious surface area surface temperature in Pretoria, South Africa. *Geocarto Int*. 2022;37(25):9707–30. <https://doi.org/10.1080/10106049.2021.2022018>.
53. Hidalgo-García D, Arco-Díaz J. Modeling the surface urban heat island (*suhi*) to study of its relationship with variations in the thermal field and with the indices of land use in the metropolitan area of Granada (spain). *Sustain Cities Soc*. 2022;87:104166. <https://doi.org/10.1016/j.scs.2022.104166>.
54. Kafy A-A, Abdullah-Al-Faisal, Islam MS, Al Rakib A, Islam MA, Khan MHH, et al. Prediction of seasonal urban thermal field variance index using machine learning algorithms in cumilla, bangladesh. *Sustain Cities Soc*. 2021;64:102542. <https://doi.org/10.1016/j.scs.2020.102542>.
55. Erdem Okumus D, Terzi F. Evaluating the role of urban fabric on surface urban heat island (*suhi*): The case of istanbul. *Sustain Cities Soc*. 2021;73:103128. <https://doi.org/10.1016/j.scs.2021.103128>.
56. Oliveira A, Lopes A, Niza S, Soares A. An urban energy balance-guided machine learning approach for synthetic nocturnal surface urban heat island prediction: A heatwave event in naples with t_s estimations. *Sci Total Environ*. 2022;805:150130. <https://doi.org/10.1016/j.scitotenv.2021.150130>.
57. Tepanosyan G, Muradyan V, Hovsepian A, Pinigin G, Medvedev A, Asmaryan S. Studying spatial-temporal changes and relationship of land cover and surface urban heat island derived through remote sensing in Yerevan, Armenia. *Build Environ*. 2021;187:107390. <https://doi.org/10.1016/j.buildenv.2020.107390>.
58. Xu S, Cheng J, Zhang Q. A random forest-based data fusion method for obtaining all-weather land surface temperature with high spatial resolution. *Remote Sensing*. 2021;13(11):2211. <https://doi.org/10.3390/rs13112211>.
59. Bansal P, Quan SJ. Examining temporally varying nonlinear effects of urban form on urban heat island using explainable machine learning: A case of seoul. *Build Environ*. 2023;247:110957. <https://doi.org/10.1016/j.buildenv.2023.110957>.
60. Blunn LP, Ames F, Croad HL, Gainford A, Higgs I, Lipson M, et al. Machine learning bias correction and downscaling of urban heatwave temperature predictions from kilometre to hectometre scale. *Meteorol Appl*. 2024;31(3):2200. <https://doi.org/10.1002/met.2200>.
61. Buoi I, Sagris V, Jaagus J. Generating synthetic daily remote sensing products suitable for surface heat island and heatwaves assessments at urban scale. *Int J Environ Sci Technol*. 2023;20(8):8599–614. <https://doi.org/10.1007/s13762-022-04510-3>.
62. Chang J-H, Lam YF, Lau S-W, Wong W-K. Development of fine-scale spatiotemporal temperature forecast model with urban climatology and geomorphometry in Hong Kong. *Urban Clim*. 2021;37:100816. <https://doi.org/10.1016/j.uclim.2021.100816>.
63. Flückiger B, Kloog I, Ragetti MS, Eeftens M, Roosli M, De Hoogh K. Modelling daily air temperature at a fine spatial resolution dealing with challenging meteorological phenomena and topography in Switzerland. *Int J Climatol*. 2022;42(12):6413–28. <https://doi.org/10.1002/joc.7597>.
64. Hu Y, Zhou X, Zhang Y, Shi S. Subpixel temperature estimation by information transfer with adaptive ensemble extreme learning machine (it-aeelm). *IEEE J Sel Top Appl Earth Observ Remote Sens*. 2021;14:6743–54. https://doi.org/10.1109/JSTA_RS.2021.3091125.

65. Jato-Espino D, Manchado C, Roldán-Valcarce A, Moscardó V. Arcuhi: a gis add-in for automated modelling of the urban heat island effect through machine learning. *Urban Clim*. 2022;44:101203. <https://doi.org/10.1016/j.uclim.2022.101203>.
66. Lemoine-Rodríguez R, Inostroza L, Zepp H. Does urban climate follow urban form? analysing intraurban 1st trajectories versus urban form trends in 3 cities with different background climates. *Sci Total Environ*. 2022;830:154570. <https://doi.org/10.1016/j.scitotenv.2022.154570>.
67. Lyu F, Wang S, Han SY, Catlett C, Wang S. An integrated CyberGIS and machine learning framework for fine-scale prediction of urban heat island using satellite remote sensing and urban sensor network data. *Urban Informatics*. 2022;1(1):6. <https://doi.org/10.1007/s44212-022-00002-4>.
68. Mohammad P, Goswami A, Chauhan S, Nayak S. Machine learning algorithm based prediction of land use land cover and land surface temperature changes to characterize the surface urban heat island phenomena over Ahmedabad city, India. *Urban Climate*. 2022;42:101116. <https://doi.org/10.1016/j.uclim.2022.101116>.
69. Oukawa GY, Krecl P, Targino AC. Fine-scale modeling of the urban heat island: a comparison of multiple linear regression and random forest approaches. *Sci Total Environ*. 2022;815:152836. <https://doi.org/10.1016/j.scitotenv.2021.152836>.
70. Lau T-K, Lin T-P. Investigating the relationship between air temperature and the intensity of urban development using on-site measurement, satellite imagery and machine learning. *Sustain Cities Soc*. 2023;100:104982. <https://doi.org/10.1016/j.scs.2023.104982>.
71. Li Y, Feng Z, Li L, Li T, Guo F, Wei J, et al. Surface urban heat islands in 932 urban region agglomerations in china during the morning and before midnight: Spatial-temporal changes, drivers, and simulation. *Geocarto Int*. 2022;37(26):13689–710. <https://doi.org/10.1080/10106049.2022.2082552>.
72. Gupta N, Aithal BH. Urban land surface temperature forecasting: a data-driven approach using regression and neural network models. *Geocarto Int*. 2024;39(1):2299145. <https://doi.org/10.1080/10106049.2023.2299145>.
73. Hassani A, Santos GS, Schneider P, Castell N. Interpolation, satellite-based machine learning, or meteorological simulation? A comparison analysis for spatio-temporal mapping of mesoscale urban air temperature. *Environ Model Assess*. 2023;29(2):291–306. <https://doi.org/10.1007/s10666-023-09943-9>.
74. Koutroumanou-Kontosi K, Cartalis C, Philippopoulos K, Agathangelidis I, Polydoros A. A methodology for bridging the gap between regional- and city-scale climate simulations for the urban thermal environment. *Climate*. 2022;10(7):106. <https://doi.org/10.3390/cli10070106>.
75. Li K, Chen Y, Jiang J. Grading surface urban heat island and investigating factor weight based on interpretable deep learning model across global cities. *Environ Int*. 2023;180:108196. <https://doi.org/10.1016/j.envint.2023.108196>.
76. Maithani S, Nautiyal G, Sharma A, Sharma SK. Simulation of land surface temperature patterns over future urban areas-a machine learning approach. *J Indian Soc Remote Sens*. 2022;50(11):2145–62. <https://doi.org/10.1007/s12524-022-01590-z>.
77. Rao P, Tassinari P, Torreggiani D. Exploring the land-use urban heat island nexus under climate change conditions using machine learning approach: A spatio-temporal analysis of remotely sensed data. *Heliyon*. 2023;9(8):18423. <https://doi.org/10.1016/j.heliyon.2023.e18423>.
78. Shao L, Liao W, Li P, Luo M, Xiong X, Liu X. Drivers of global surface urban heat islands: Surface property, climate background, and 2d/3d urban morphologies. *Build Environ*. 2023;242:110581. <https://doi.org/10.1016/j.buildenv.2023.110581>.
79. Shang K, Xu L, Liu X, Yin Z, Liu Z, Li X, et al. Study of urban heat island effect in hangzhou metropolitan area based on swtles algorithm and image dichotomous model. *SAGE Open*. 2023;13(4):21582440231208852. <https://doi.org/10.1177/21582440231208852>.
80. Varentsov M, Krinitskiy M, Stepanenko V. Machine learning for simulation of urban heat island dynamics based on large-scale meteorological conditions. *Climate*. 2023;11(10):200. <https://doi.org/10.3390/cli11100200>.
81. Vergara DCDM, Blanco AC, Marciano JJS, Meneses SF, Borlongan NJB, Sabuito AJC. Assessing and modelling urban heat island in baguio city using landsat imagery and machine learning. In: *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLVIII-4/W6-2022. 2023. pp. 457–464. <https://doi.org/10.5194/isprs-archives-XLVIII-4-W6-2022-457-2023>.
82. Bindajam AA, Mallick J, Talukdar S, Shahfahad Shohan AAA, Rahman A. Modeling the spatiotemporal heterogeneity of land surface temperature and its relationship with land use land cover using geo-statistical techniques and machine learning algorithms. *Environ Sci Pollut Res*. 2022;30(49):106917–35. <https://doi.org/10.1007/s11356-022-23211-5>.
83. Ding X, Zhao Y, Fan Y, Li Y, Ge J. Machine learning-assisted mapping of city-scale air temperature: Using sparse meteorological data for urban climate modeling and adaptation. *Build Environ*. 2023;234:110211. <https://doi.org/10.1016/j.buildenv.2023.110211>.
84. Mansourmoghaddam M, Roustai I, Ghafarian Malamiri H, Sadeghnejad M, Krzyszczak J, Ferreira CSS. Modeling and estimating the land surface temperature (lst) using remote sensing and machine learning (case study: Yazd, Iran). *Remote Sens*. 2024;16(3):454. <https://doi.org/10.3390/rs16030454>.
85. Oukawa GY, Krecl P, Targino AC, Batista LFA. Advantages of modeling the urban heat island intensity: a tool for implementing nature-based solutions. *Sustain Cities Soc*. 2024;102:105204. <https://doi.org/10.1016/j.scs.2024.105204>.
86. Zhao Z, Li H, Wang S. Machine learning-based surrogate models for fast impact assessment of a new building on urban local microclimate at design stage. *Build Environ*. 2024;266:112142. <https://doi.org/10.1016/j.buildenv.2024.112142>.
87. Du Y, Xie Z, Zhang L, Wang N, Wang M, Hu J. Machine-learning-assisted characterization of regional heat islands with a spatial extent larger than the urban size. *Remote Sensing*. 2024;16(3):599. <https://doi.org/10.3390/rs16030599>.
88. Liu X, Liu Z, Zhu Z, Qiao R. Decoupling the multi-drivers of urban extreme heat environment in urban agglomerations using ensemble learning. *Build Environ*. 2024;258:111618. <https://doi.org/10.1016/j.buildenv.2024.111618>.
89. Kavathekar V, Tripathy AK, Chettri SK, Bhanage V. Evaluation of land use land cover dynamics and urban heat island effects over Mumbai metropolitan region, India. *Int J Environ Sci Technol*. 2024. <https://doi.org/10.1007/s13762-024-06312-1>.
90. Mendez-Astudillo J, Mendez-Astudillo M. A machine learning approach to monitoring the UHI from GNSS data. *IEEE Trans Geosci Remote Sens*. 2022;60:1–11. <https://doi.org/10.1109/TGRS.2021.3091949>.
91. Huo X, Cui G, Ma L, Tang B, Tang R, Shao K, et al. Urban land surface temperature prediction using parallel stl-bi-lstm neural network. *J Appl Remote Sens*. 2022;16(3):034529–034529. <https://doi.org/10.1117/1.JRS.16.034529>.
92. Choe Y-J, Yom J-H. Improving accuracy of land surface temperature prediction model based on deep-learning. *Spat Inf Res*. 2019;28(3):377–82. <https://doi.org/10.1007/s41324-019-00299-5>.

93. Cho D, Yoo C, Son B, Im J, Yoon D, Cha D-H. A novel ensemble learning for post-processing of nwp model's next-day maximum air temperature forecast in summer using deep learning and statistical approaches. *Weather Clim Extremes*. 2022;35:100410. <https://doi.org/10.1016/j.wace.2022.100410>.
94. Javanroodi K, Nik VM, Giometto MG, Scartezzini J-L. Combining computational fluid dynamics and neural networks to characterize microclimate extremes: learning the complex interactions between meso-climate and urban morphology. *Sci Total Environ*. 2022;829:154223. <https://doi.org/10.1016/j.scitotenv.2022.154223>.
95. Khan MI, Maity R. Hybrid deep learning approach for multi-step-ahead prediction for daily maximum temperature and heatwaves. *Theor Appl Climatol*. 2022;149(3–4):945–63. <https://doi.org/10.1007/s00704-022-04103-7>.
96. Delgado-Enalesa I, Lizundia-Loiola J, Molina-Costa P, Ser JD. A machine learning approach for the efficient estimation of ground-level air temperature in urban areas. *Urban Climate*. 2023;49:101775. <https://doi.org/10.1016/j.uclim.2023.101775>.
97. Johannsen F, Soares PMM, Langendijk GS. On the deep learning approach for improving the representation of urban climate: the paris urban heat island and temperature extremes. *Urban Clim*. 2024;56:102039. <https://doi.org/10.1016/j.uclim.2024.102039>.
98. Chen L, Fang B, Zhao L, Zang Y, Liu W, Chen Y, et al. Deepurbandownscale: a physics-informed deep learning framework for high-resolution urban surface temperature estimation via 3d point clouds. *Int J Appl Earth Obs Geoinf*. 2022;106:102650. <https://doi.org/10.1016/j.jag.2021.102650>.
99. Zhang X, Chen W, Chen Z, Yang F, Meng C, Gou P, et al. Construction of cloud-free modis-like land surface temperatures coupled with a regional weather research and forecasting (wrf) model. *Atmos Environ*. 2022;283:119190. <https://doi.org/10.1016/j.atmosenv.2022.119190>.
100. Zhu H-C, Ren C, Wang J, Feng Z, Haghighat F, Cao S-J. Fast prediction of spatial temperature distributions in urban areas with *wrf* and temporal fusion transformers (*tft*). *Sustain Cities Soc*. 2024;103:105249. <https://doi.org/10.1016/j.scs.2024.105249>.
101. Wu Y, Teufel B, Sushama L, Belair S, Sun L. Deep learning-based super-resolution climate simulator-emulator framework for urban heat studies at 10^{-3} scale. *Geophys Res Lett*. 2021;48(19):2021–094737. <https://doi.org/10.1029/2021GL094737>.
102. Chajaei F, Bagheri H. Machine learning framework for high-resolution air temperature downscaling using lidar-derived urban morphological features. *Urban Climate*. 2024;57:102102. <https://doi.org/10.1016/j.uclim.2024.102102>.
103. Sundar G, Patchaammal P, Venkateshwarulu BV, Kumar TP, Rajamannar K, Tripathi RK. Integrative remote sensing approaches using generative adversarial networks for urban heat island analysis and mitigation. *Remote Sens Earth Syst Sci*. 2024;7(4):681–98. <https://doi.org/10.1007/s41976-024-00156-6>.
104. Kong J, Ryu Y, Jeong S, Zhong Z, Choi W, Kim J, et al. Long-term vegetation monitoring. *ISPRS J Photogramm Remote Sens*. 2023;200:1–23. <https://doi.org/10.1016/j.isprsjprs.2023.04.013>.
105. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine learning in python. *J Mach Learn Res*. 2011;12:2825–30.
106. Abadi M, Barham P, Chen J, Chen Z, Davis A, Dean J, Devin M, Ghemawat S, Irving G, Isard M, Kudlur M, Levenberg J, Monga R, Moore S, Murray DG, Steiner B, Tucker P, Vasudevan V, Warden P, Wicke M, Yu Y, Zheng X. Tensorflow: A system for large-scale machine learning. In: *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI 2016)*, pp. 265–283. USENIX Association, Savannah, GA, USA 2016.
107. Hawick KA, Coddington P, James D. Distributed frameworks and parallel algorithms for processing large-scale geographic data. *Int J Geogr Inf Sci*. 2003;17(3):251–72. <https://doi.org/10.1080/13658810210157777>.
108. Provost F, Fawcett T, Kohavi R. The case against accuracy estimation for comparing induction algorithms. In: *Proceedings of the Fifteenth International Conference on Machine Learning (ICML)*, pp. 445–453. Morgan Kaufmann, San Francisco, CA 1998.
109. Willmott CJ, Matsuura K. Advantages of the mean absolute error (mae) over the root mean square error (rmse) in assessing average model performance. *Clim Res*. 2005;30(1):79–82. <https://doi.org/10.3354/cr030079>.
110. Chai T, Draxler RR. Root mean square error (*rmse*) or mean absolute error (*mae*)? – arguments against avoiding *rmse* in the literature. *Geosci Model Dev*. 2014;7(3):1247–50. <https://doi.org/10.5194/gmd-7-1247-2014>.
111. Taylor J, Wilkinson P, Davies M, Armstrong B, Chalabi Z, Mavrogianni A, et al. Mapping the effects of urban heat island, housing, and age on excess heat-related mortality in London. *Urban Climate*. 2015;14:517–28. <https://doi.org/10.1016/j.uclim.2015.08.001>.
112. Pande CB, Egbueri JC, Costache R, Sidek LM, Wang Q, Alshehri F, et al. Predictive modeling of land surface temperature (lst) based on landsat-8 satellite data and machine learning models for sustainable development. *J Clean Prod*. 2024;444:141035. <https://doi.org/10.1016/j.jclepro.2024.141035>.
113. Doshi-Velez F, Kim B. Towards a rigorous science of interpretable machine learning. 2017. [arXiv:1702.08608](https://arxiv.org/abs/1702.08608).
114. Ribeiro MT, Singh S, Guestrin C. “why should i trust you?”: Explaining the predictions of any classifier 2016. <https://doi.org/10.48550/arXiv.1602.04938><https://arxiv.org/abs/1602.04938>.
115. Lundberg SM, Lee S-I. A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pp. 4765–4774. Curran Associates, Inc., Long Beach, CA, USA 2017. <https://doi.org/10.48550/arXiv.1705.07874>.
116. Craven MW, Shavlik JW. Extracting tree-structured representations of trained networks. In: *Proceedings of the 8th International Conference on Neural Information Processing Systems*, pp. 24–30. MIT Press, Cambridge, MA, USA 1996. <https://doi.org/10.5555/2998828.2998832>.
117. Wachter S, Mittelstadt B, Russell C. Counterfactual explanations without opening the black box: Automated decisions and the gdpr. *Harv JL & Tech*. 2018;31(2):841–87. <https://doi.org/10.2139/ssrn.3063289>.
118. Addas A. Machine learning techniques to map the impact of urban heat island: investigating the city of Jeddah. *Land*. 2023;12(6):1159. <https://doi.org/10.3390/land12061159>.